

BIOLOGY

TENTH EDITION

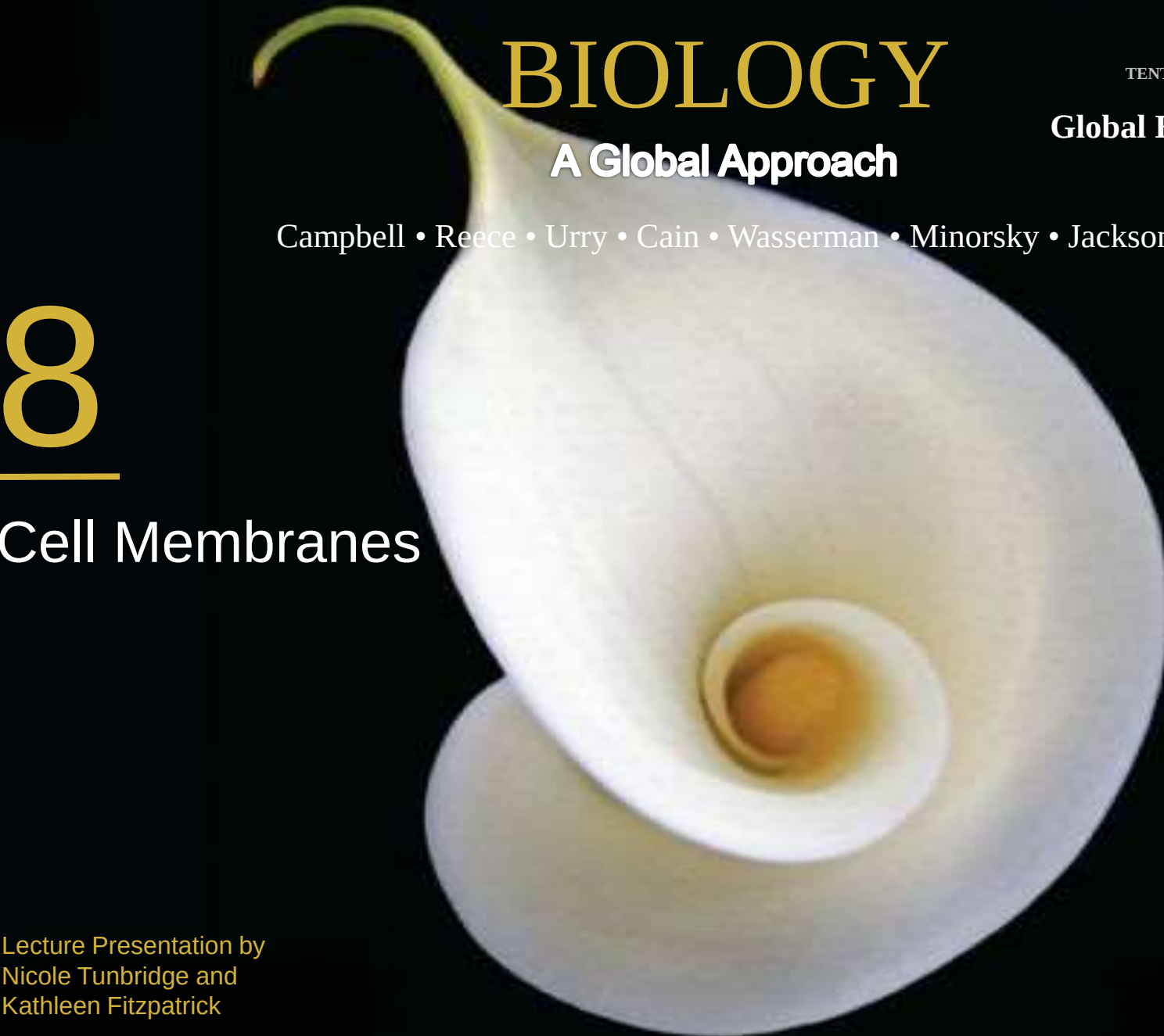
Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson

8

Cell Membranes



Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

Life at the Edge

- a) The plasma membrane is the boundary that separates the living cell from its surroundings
- b) The plasma membrane exhibits **selective permeability**, allowing some substances to cross it more easily than others

Figure 7.1

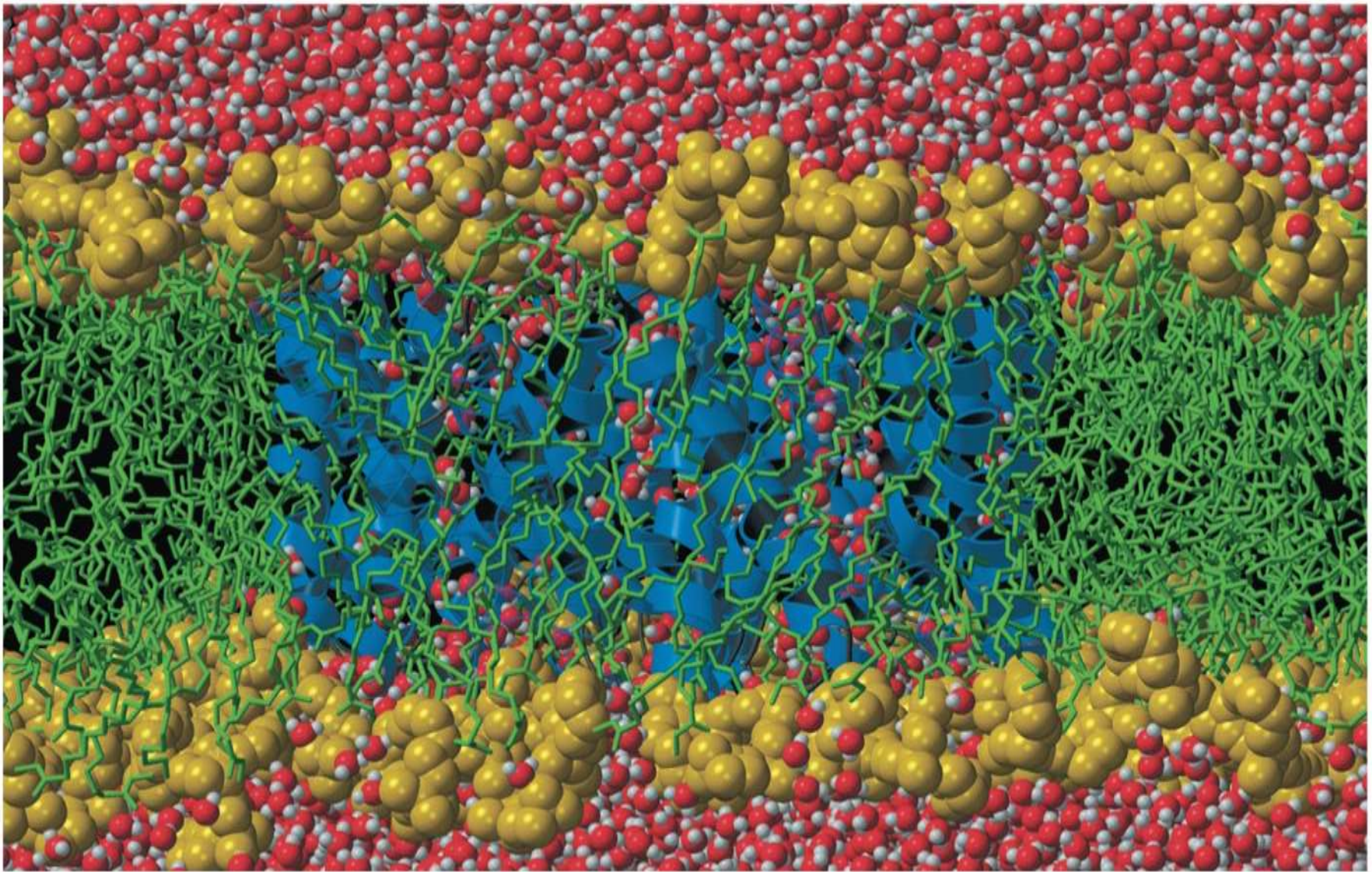
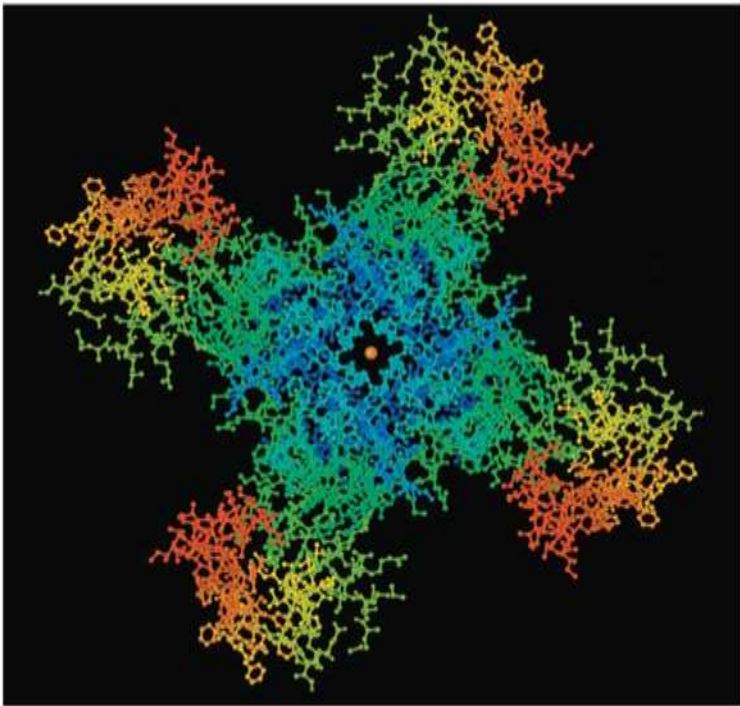


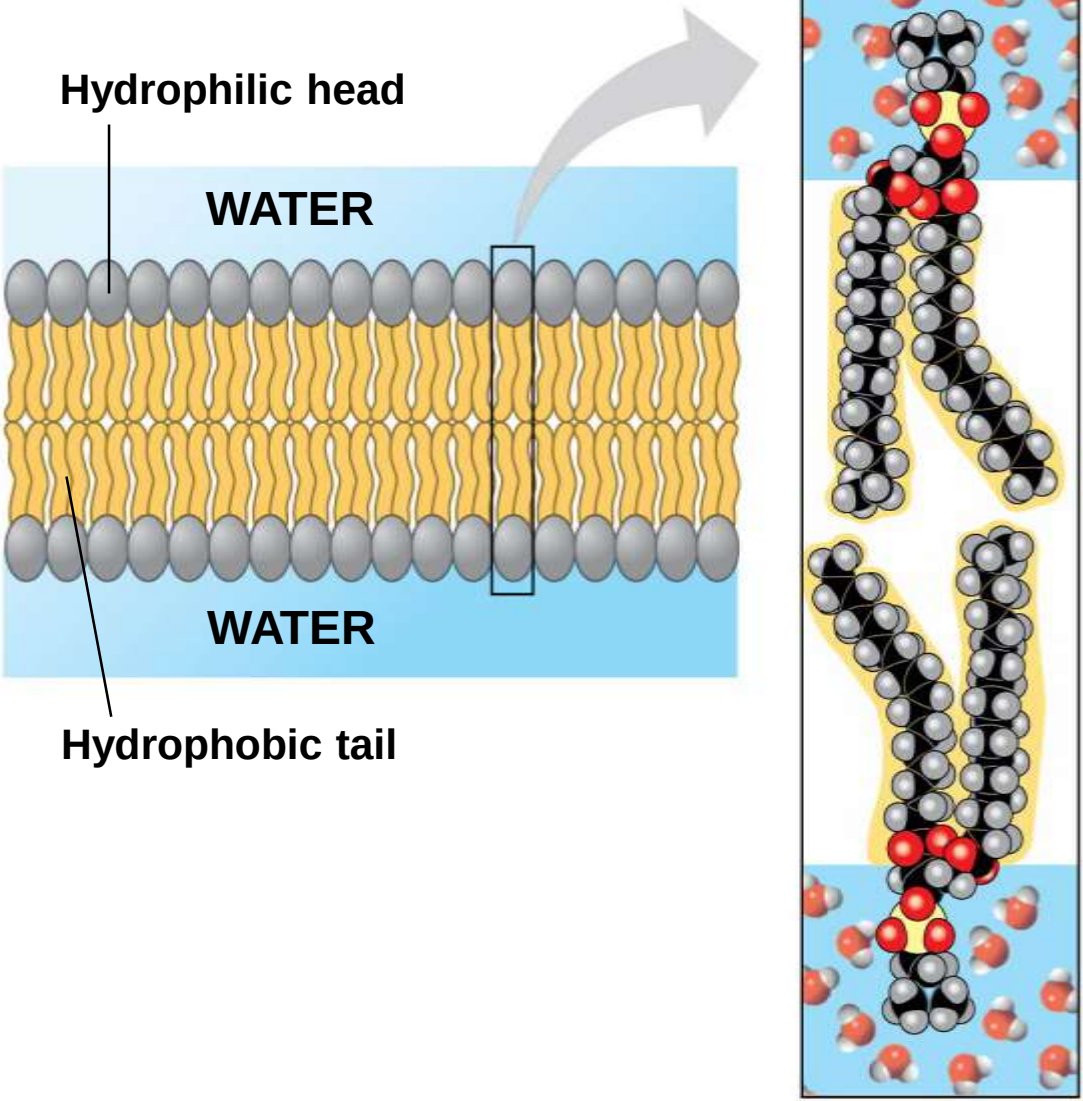
Figure 7.1a



Concept 7.1: Cellular membranes are fluid mosaics of lipids and proteins

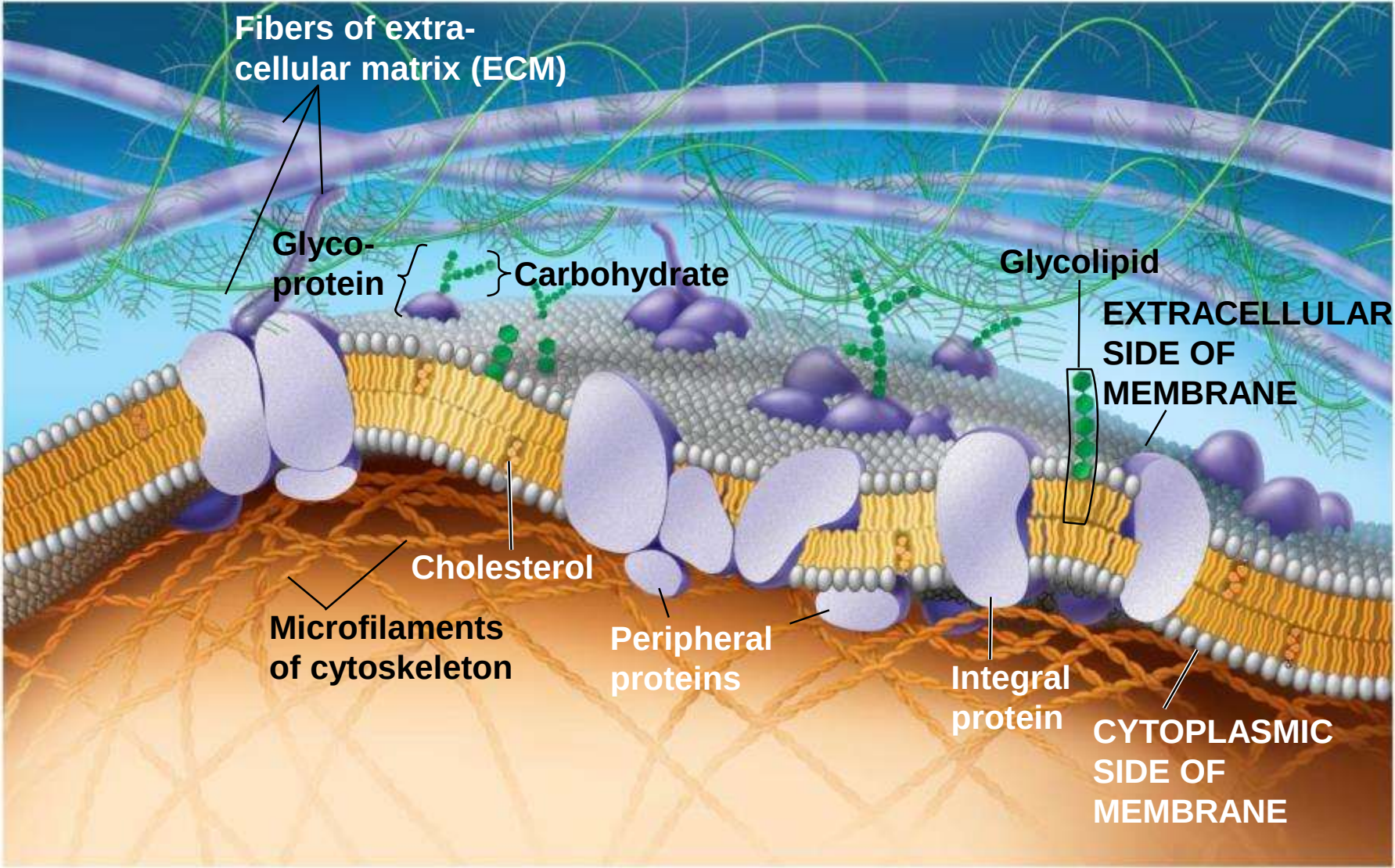
- a) Phospholipids are the most abundant lipid in the plasma membrane
- b) Phospholipids are **amphipathic** molecules, containing hydrophobic and hydrophilic regions
- c) A phospholipid bilayer can exist as a stable boundary between two aqueous compartments

Figure 7.2



- a) The **fluid mosaic model** states that a membrane is a fluid structure with a “mosaic” of various proteins embedded in it
- b) Proteins are not randomly distributed in the membrane

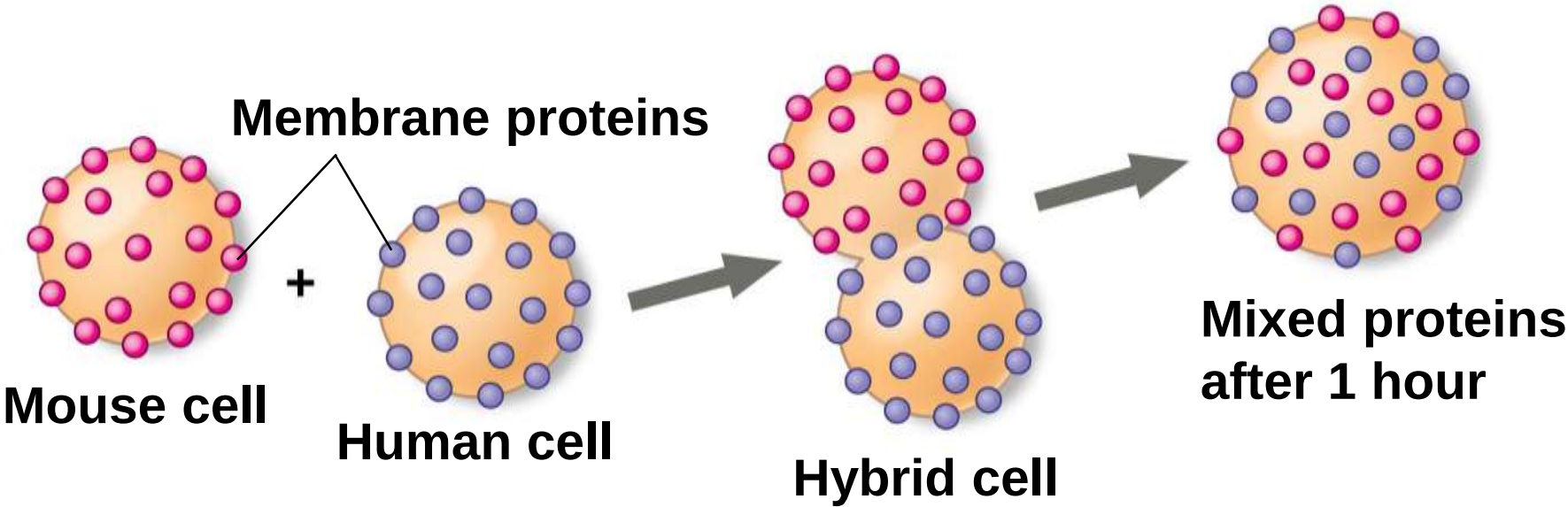
Figure 7.3



The Fluidity of Membranes

- a) Phospholipids in the plasma membrane can move within the bilayer
- b) Most of the lipids, and some proteins, drift laterally
- c) Rarely, a lipid may flip-flop transversely across the membrane

Figure 7.4-3

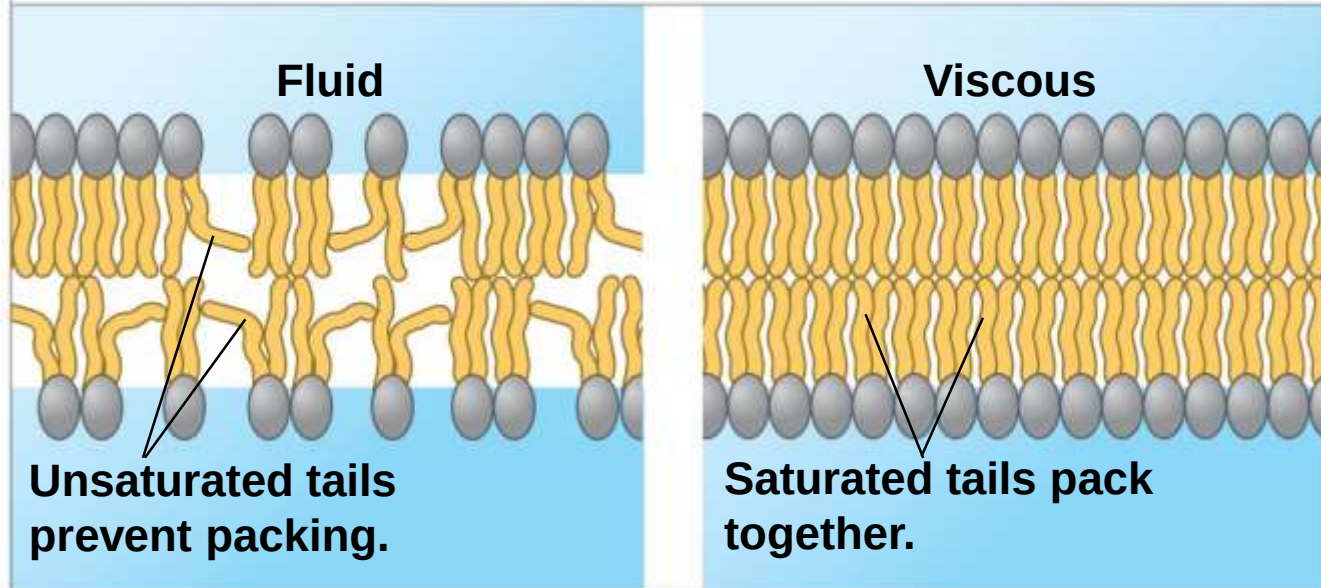


- a) As temperatures cool, membranes switch from a fluid state to a solid state
- b) The temperature at which a membrane solidifies depends on the types of lipids
- c) Membranes rich in unsaturated fatty acids are more fluid than those rich in saturated fatty acids
- d) Membranes must be fluid to work properly; they are usually about as fluid as salad oil

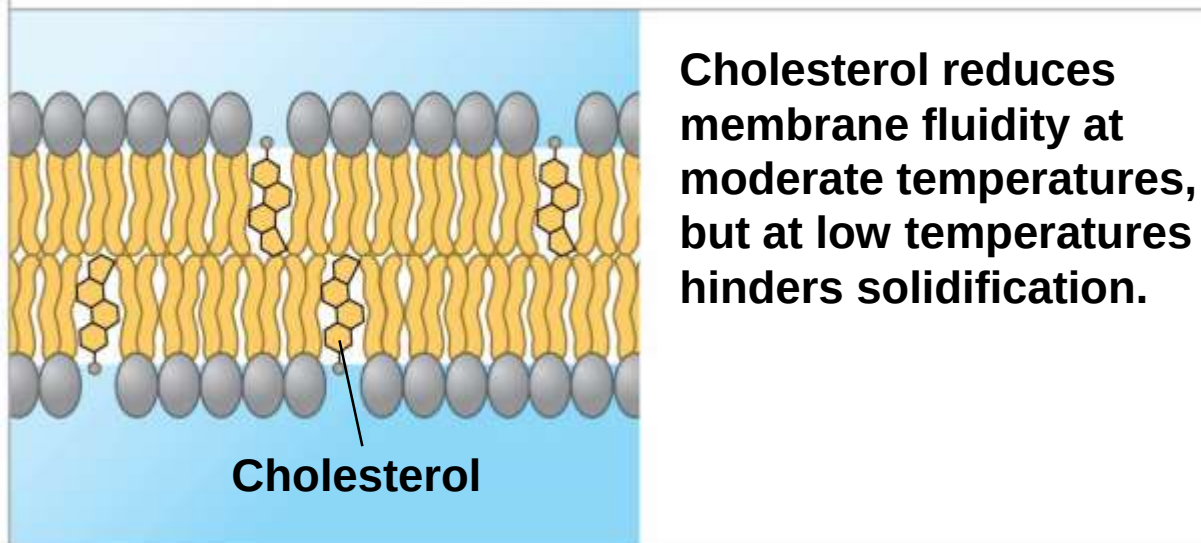
- a) The steroid cholesterol has different effects on membrane fluidity at different temperatures
- b) At warm temperatures (such as 37°C), cholesterol restrains movement of phospholipids
- c) At cool temperatures, it maintains fluidity by preventing tight packing

Figure 7.5

(a) Unsaturated versus saturated hydrocarbon tails



(b) Cholesterol within the animal cell membrane



Evolution of Differences in Membrane Lipid Composition

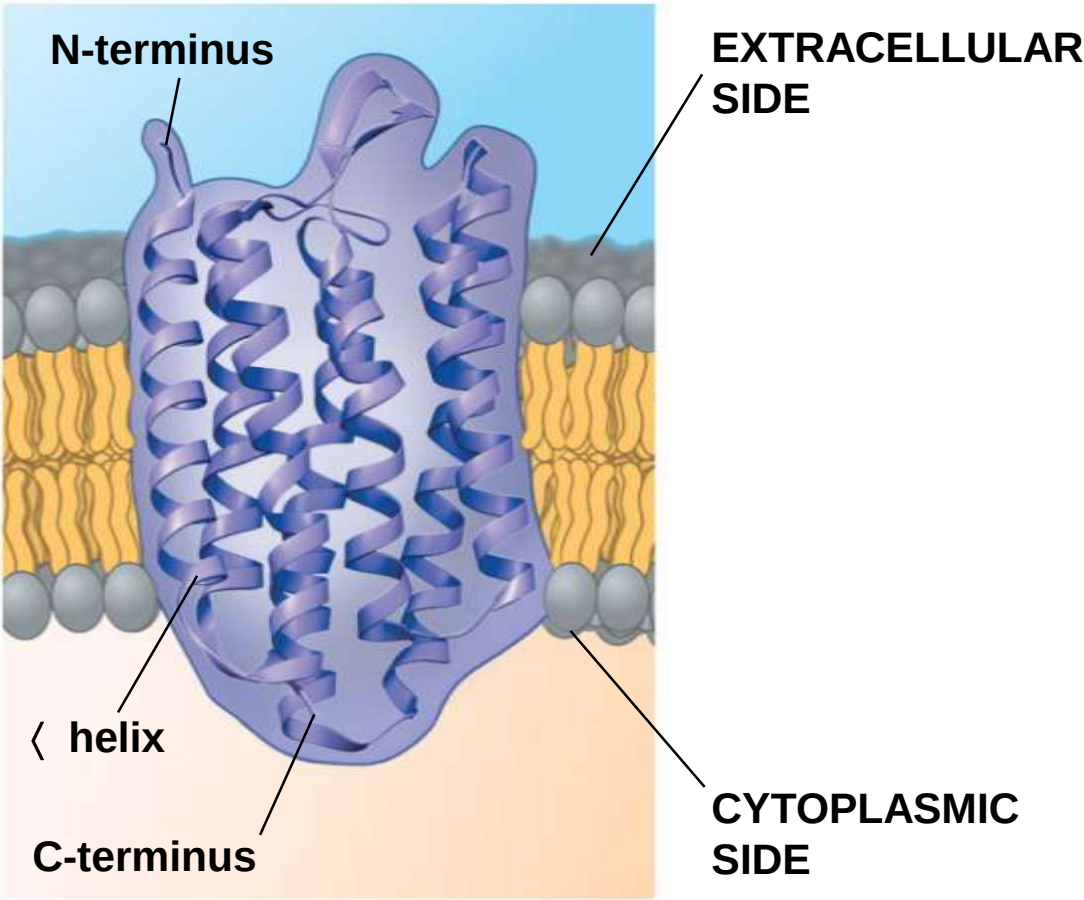
- a) Variations in lipid composition of cell membranes of many species appear to be adaptations to specific environmental conditions
- b) Ability to change the lipid compositions in response to temperature changes has evolved in organisms that live where temperatures vary

Membrane Proteins and Their Functions

- a) A membrane is a collage of different proteins, often grouped together, embedded in the fluid matrix of the lipid bilayer
- b) Proteins determine most of the membrane's specific functions

- a) **Peripheral proteins** are bound to the surface of the membrane
- b) **Integral proteins** penetrate the hydrophobic core
- c) Integral proteins that span the membrane are called transmembrane proteins
- d) The hydrophobic regions of an integral protein consist of one or more stretches of nonpolar amino acids, often coiled into alpha helices

Figure 7.6



a) Six major functions of membrane proteins

a) Transport

b) Enzymatic activity

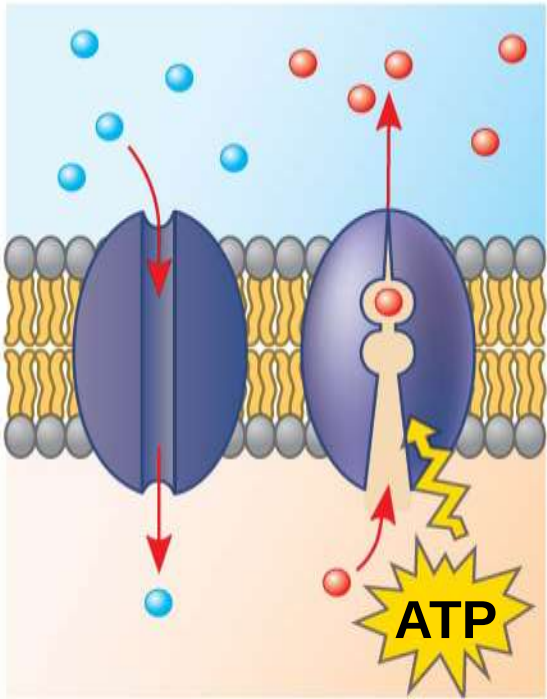
c) Signal transduction

d) Cell-cell recognition

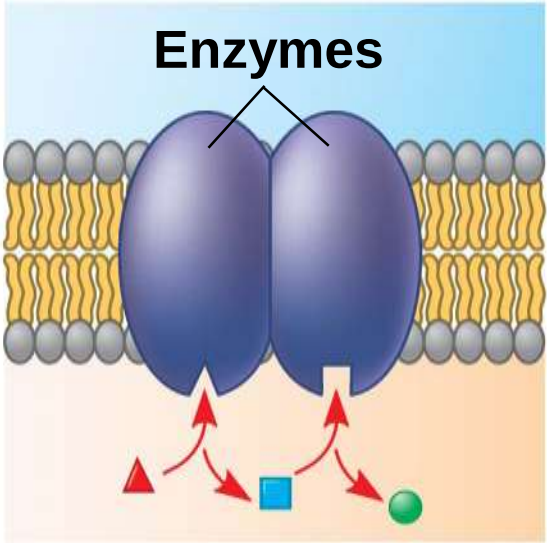
e) Intercellular joining

f) Attachment to the cytoskeleton and extracellular matrix (ECM)

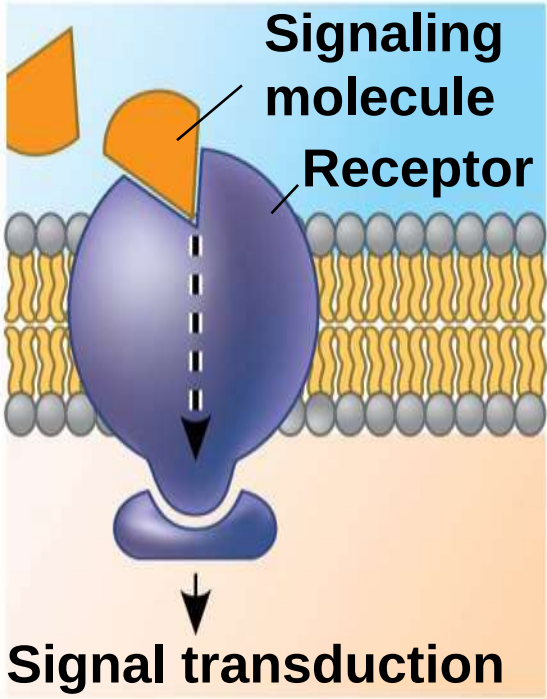
Figure 7.7a



(a) Transport

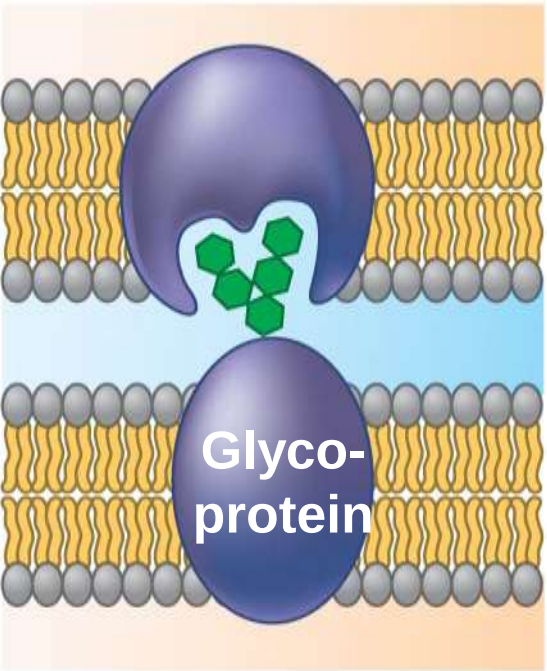


(b) Enzymatic activity

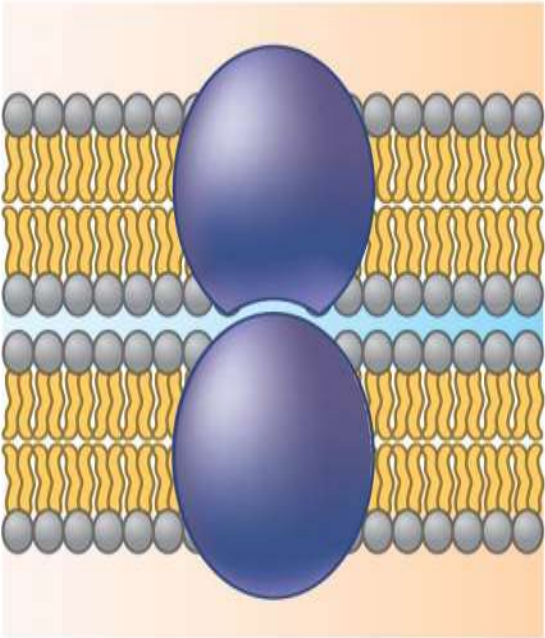


(c) Signal transduction

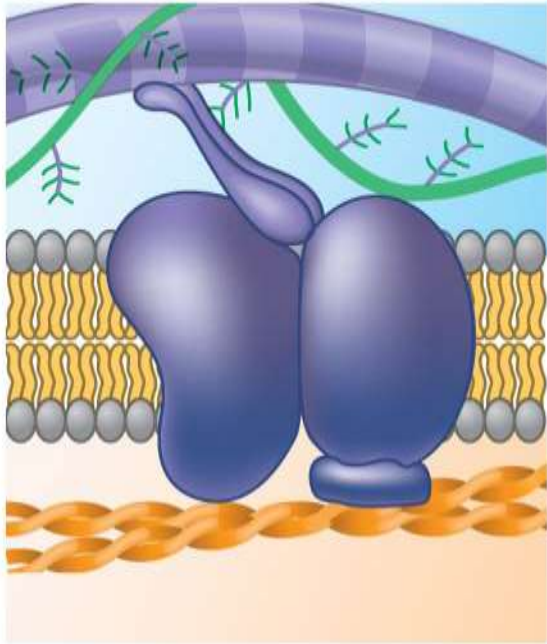
Figure 7.7b



(d) Cell-cell recognition



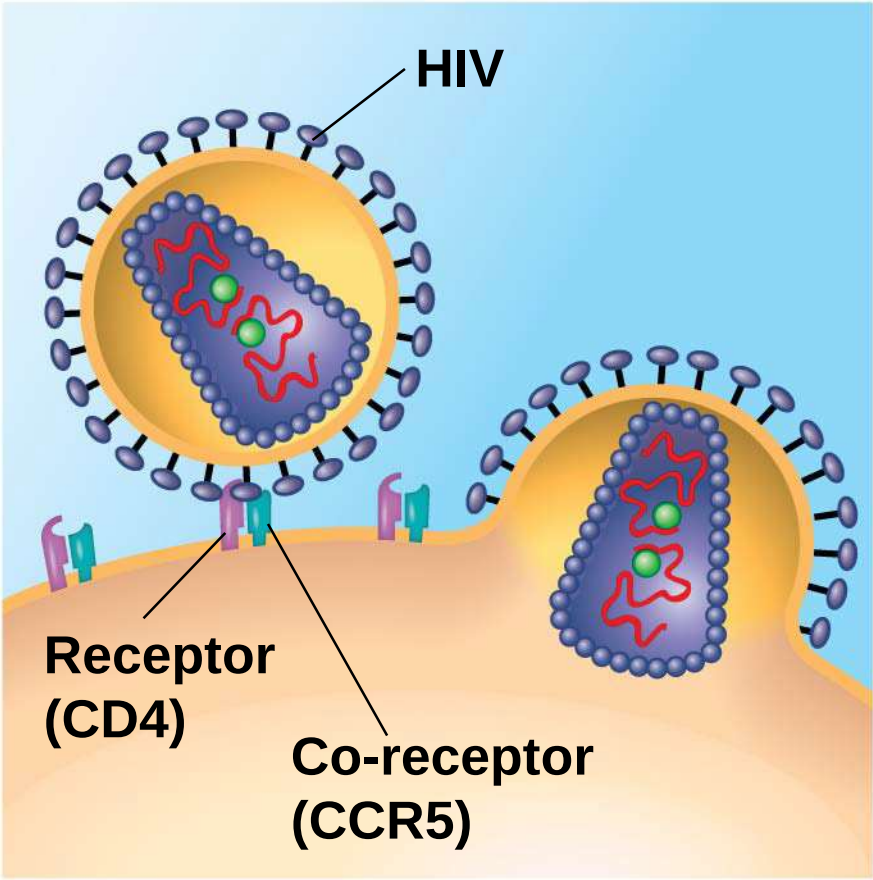
(e) Intercellular joining



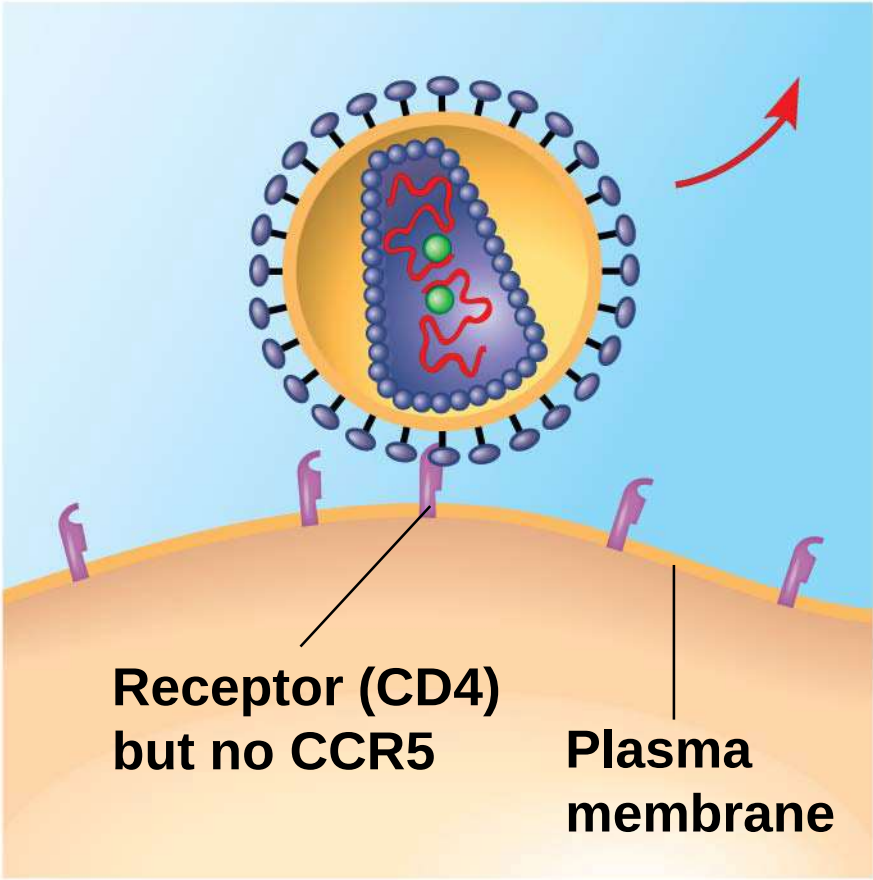
(f) Attachment to the cytoskeleton and extracellular matrix (ECM)

- a) HIV must bind to the immune cell surface protein CD4 and a “co-receptor” CCR5 in order to infect a cell
- b) HIV cannot enter the cells of resistant individuals that lack CCR5

Figure 7.8



(a)



(b)

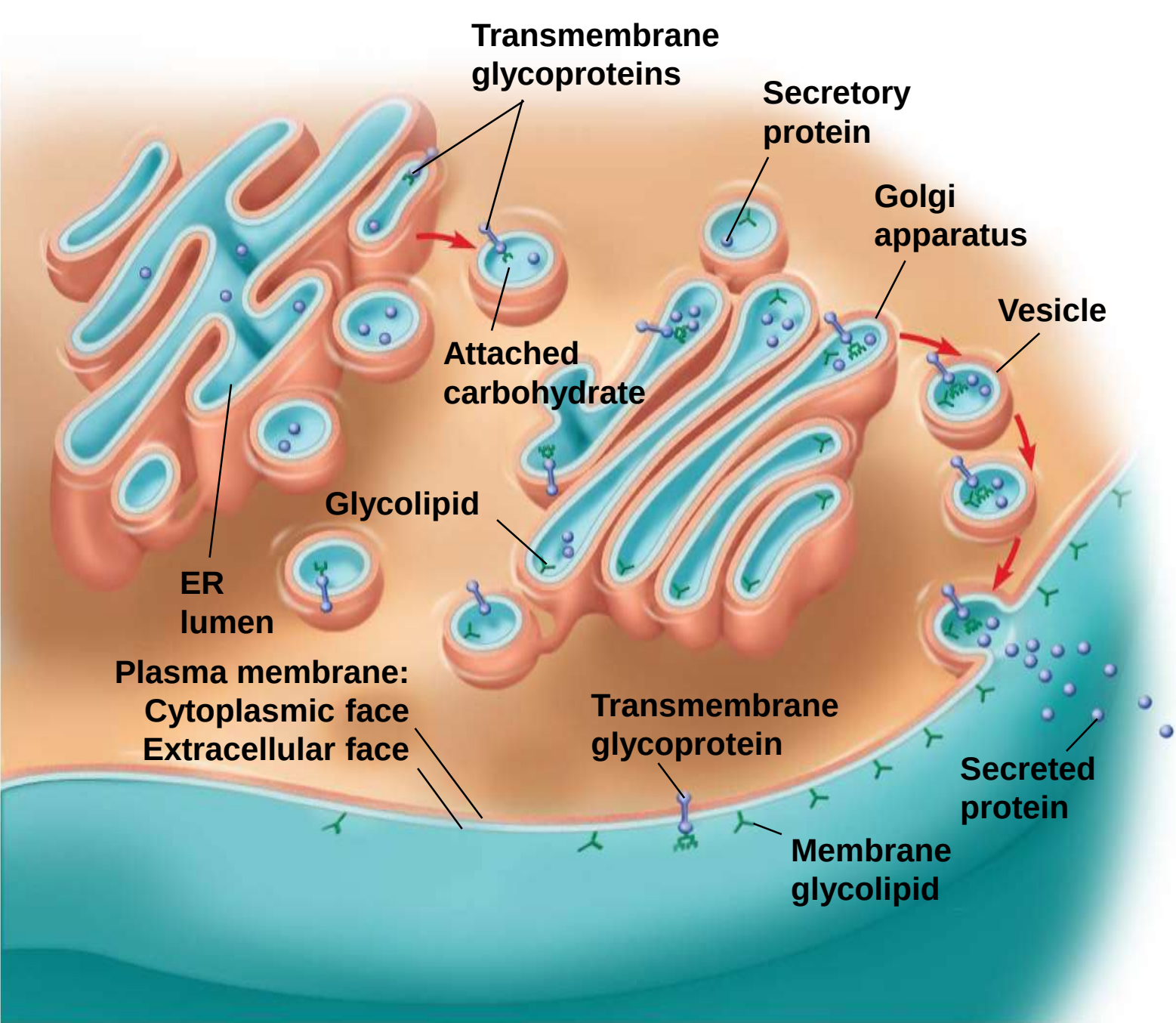
The Role of Membrane Carbohydrates in Cell-Cell Recognition

- a) Cells recognize each other by binding to molecules, often containing carbohydrates, on the extracellular surface of the plasma membrane
- b) Membrane carbohydrates may be covalently bonded to lipids (forming **glycolipids**) or more commonly to proteins (forming **glycoproteins**)
- c) Carbohydrates on the external side of the plasma membrane vary among species, individuals, and even cell types in an individual

Synthesis and Sidedness of Membranes

- a) Membranes have distinct inside and outside faces
- b) The asymmetrical distribution of proteins, lipids, and associated carbohydrates in the plasma membrane is determined when the membrane is built by the ER and Golgi apparatus

Figure 7.9



Concept 7.2: Membrane structure results in selective permeability

- a) A cell must exchange materials with its surroundings, a process controlled by the plasma membrane
- b) Plasma membranes are selectively permeable, regulating the cell's molecular traffic

The Permeability of the Lipid Bilayer

- a) Hydrophobic (nonpolar) molecules, such as hydrocarbons, can dissolve in the lipid bilayer and pass through the membrane rapidly
- b) Hydrophilic molecules including ions and polar molecules do not cross the membrane easily

Transport Proteins

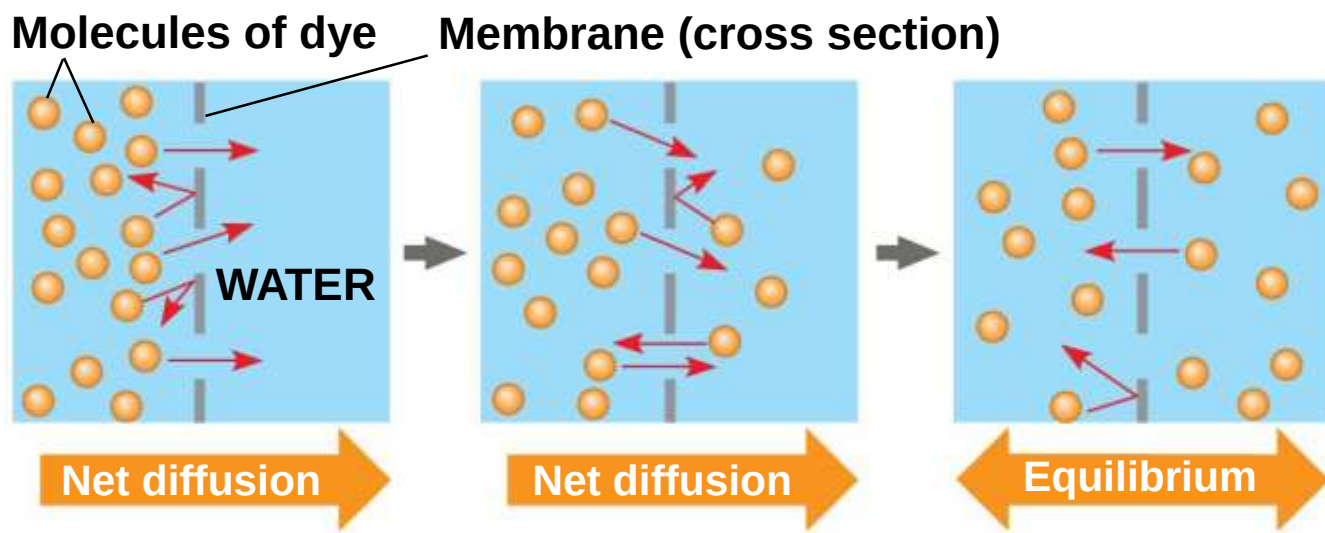
- a) **Transport proteins** allow passage of hydrophilic substances across the membrane
- b) Some transport proteins, called channel proteins, have a hydrophilic channel that certain molecules or ions can use as a tunnel
- c) Channel proteins called **aquaporins** facilitate the passage of water

- a) Other transport proteins, called carrier proteins, bind to molecules and change shape to shuttle them across the membrane
- b) A transport protein is specific for the substance it moves

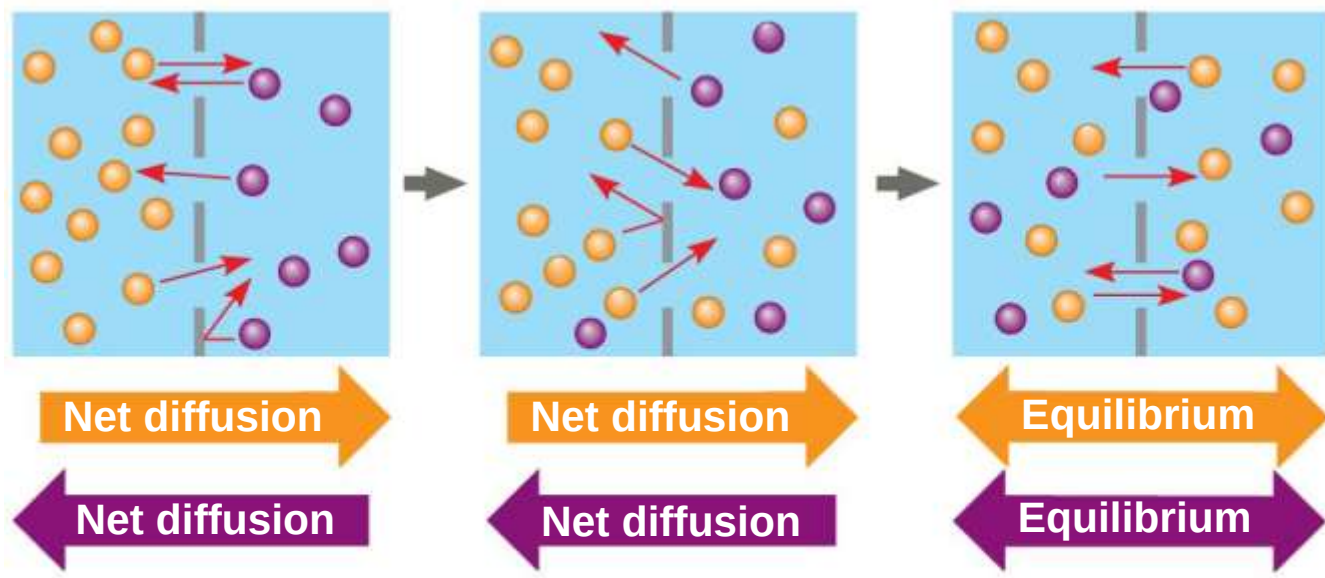
Concept 7.3: Passive transport is diffusion of a substance across a membrane with no energy investment

- a) **Diffusion** is the tendency for molecules to spread out evenly into the available space
- b) Although each molecule moves randomly, diffusion of a population of molecules may be directional
- c) At dynamic equilibrium, as many molecules cross the membrane in one direction as in the other

Figure 7.10



(a) Diffusion of one solute



(b) Diffusion of two solutes

- a) Substances diffuse down their **concentration gradient**, the region along which the density of a chemical substance increases or decreases
- b) No work must be done to move substances down the concentration gradient
- c) The diffusion of a substance across a biological membrane is **passive transport** because no energy is expended by the cell to make it happen

Effects of Osmosis on Water Balance

- a) **Osmosis** is the diffusion of water across a selectively permeable membrane
- b) Water diffuses across a membrane from the region of lower solute concentration to the region of higher solute concentration until the solute concentration is equal on both sides

Figure 7.11

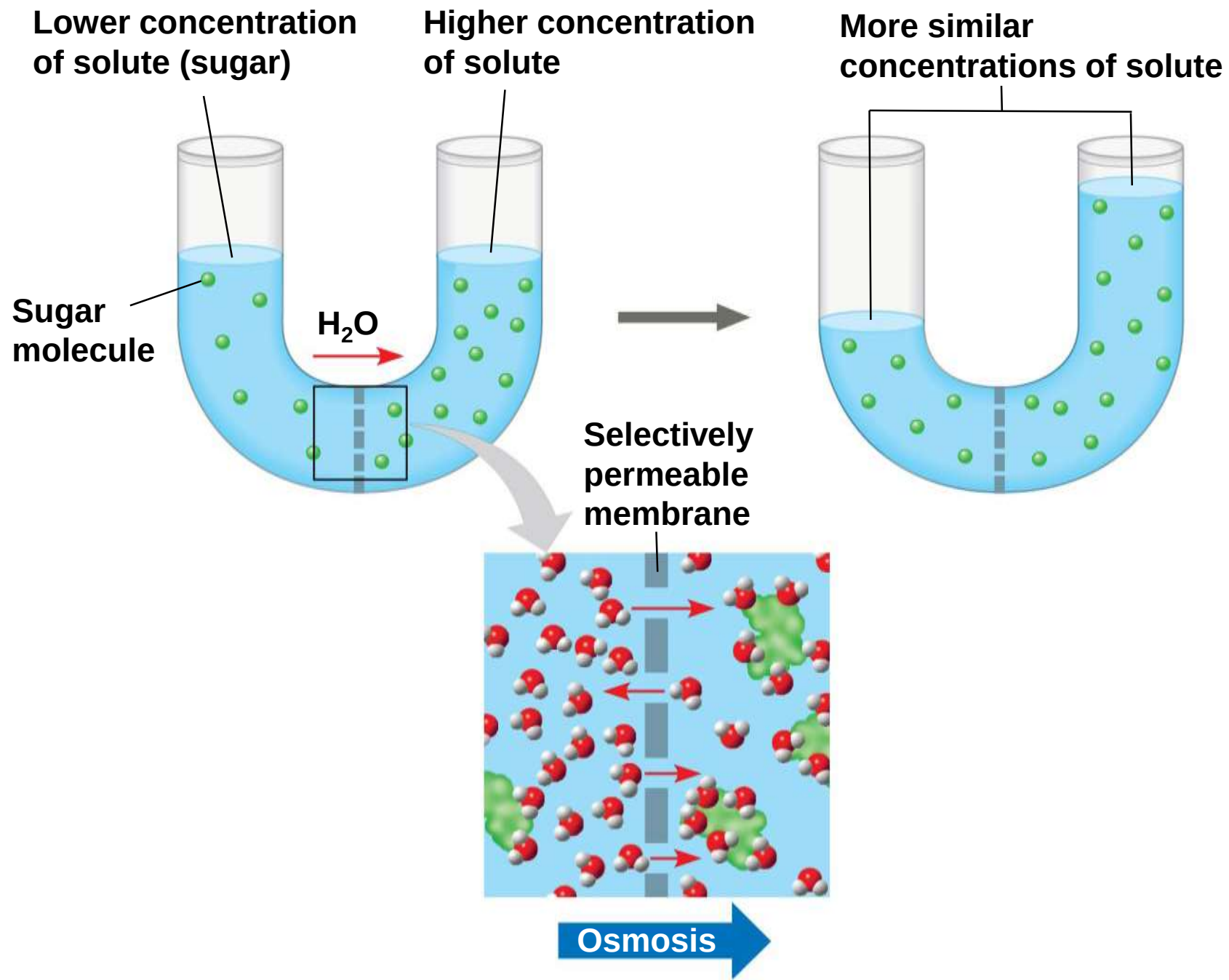
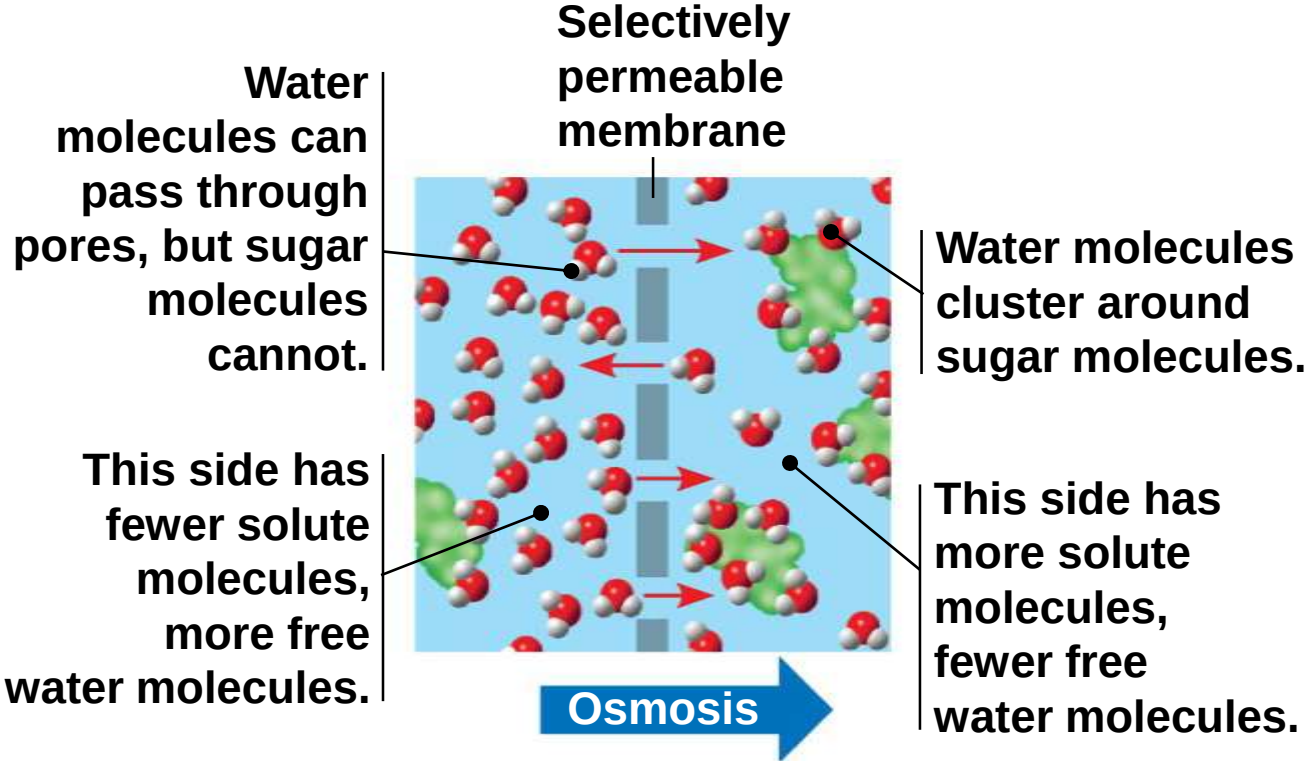


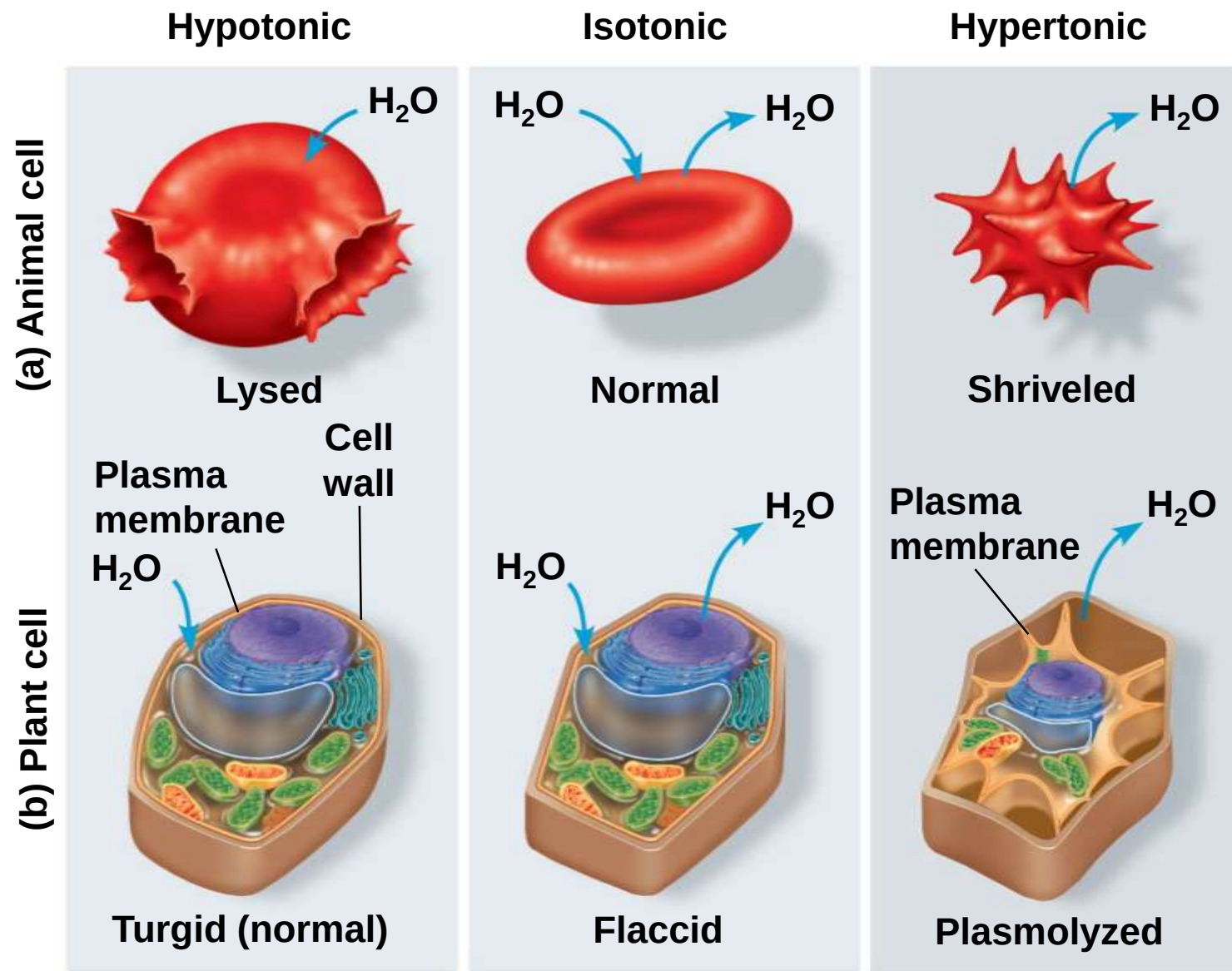
Figure 7.11a



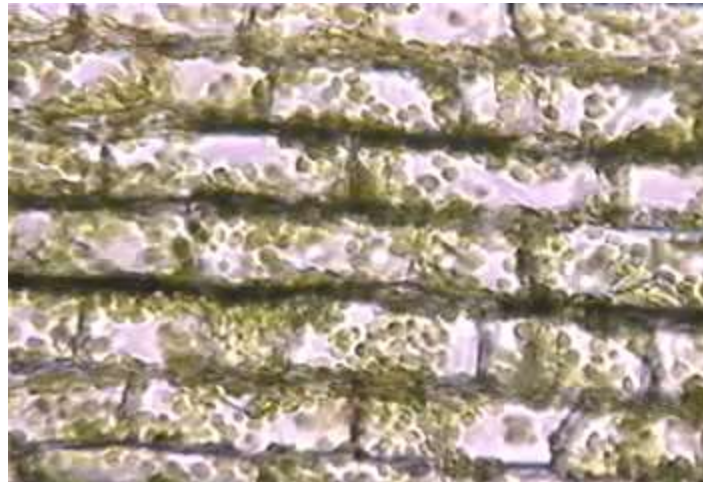
Water Balance of Cells Without Cell Walls

- a) Tonicity** is the ability of a surrounding solution to cause a cell to gain or lose water
- b) Isotonic** solution: Solute concentration is the same as that inside the cell; no *net* water movement across the plasma membrane
- c) Hypertonic** solution: Solute concentration is greater than that inside the cell; cell loses water
- d) Hypotonic** solution: Solute concentration is less than that inside the cell; cell gains water

Figure 7.12

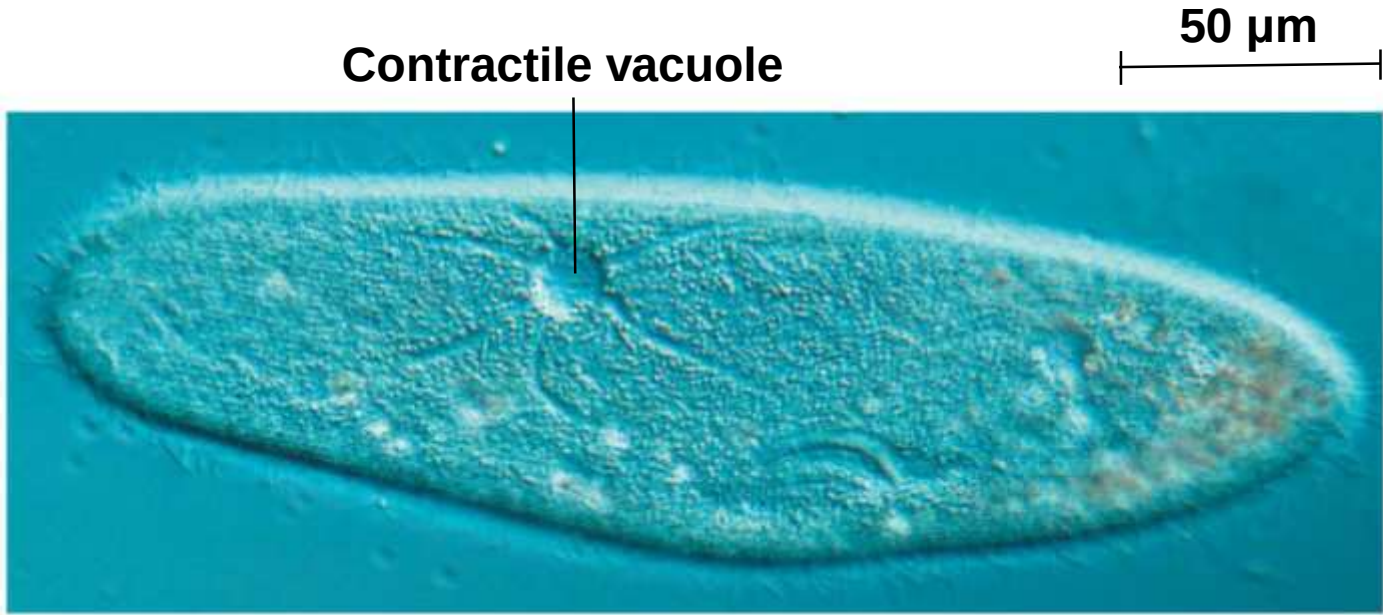


Video: Plasmolysis



- a) Hypertonic or hypotonic environments create osmotic problems for organisms
- b) Osmoregulation**, the control of solute concentrations and water balance, is a necessary adaptation for life in such environments
- c) The protist *Paramecium*, which is hypertonic to its pond water environment, has a contractile vacuole that acts as a pump

Figure 7.13



Water Balance of Cells with Cell Walls

- a) Cell walls help maintain water balance
- b) A plant cell in a hypotonic solution swells until the wall opposes uptake; the cell is now **turgid** (firm)
- c) If a plant cell and its surroundings are isotonic, there is no net movement of water into the cell; the cell becomes **flaccid** (limp)

- a) In a hypertonic environment, plant cells lose water
- b) The membrane pulls away from the cell wall causing the plant to wilt, a usually lethal effect called **plasmolysis**

Facilitated Diffusion: Passive Transport Aided by Proteins

- a) In **facilitated diffusion**, transport proteins speed the passive movement of molecules across the plasma membrane
- b) Transport proteins include channel proteins and carrier proteins

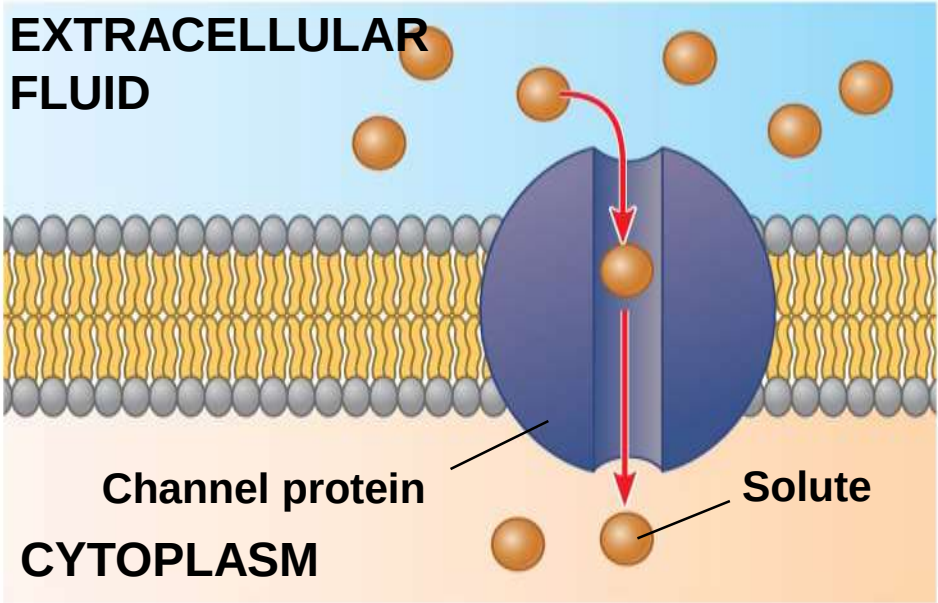
a) Channel proteins provide corridors that allow a specific molecule or ion to cross the membrane

b) Aquaporins facilitate the diffusion of water

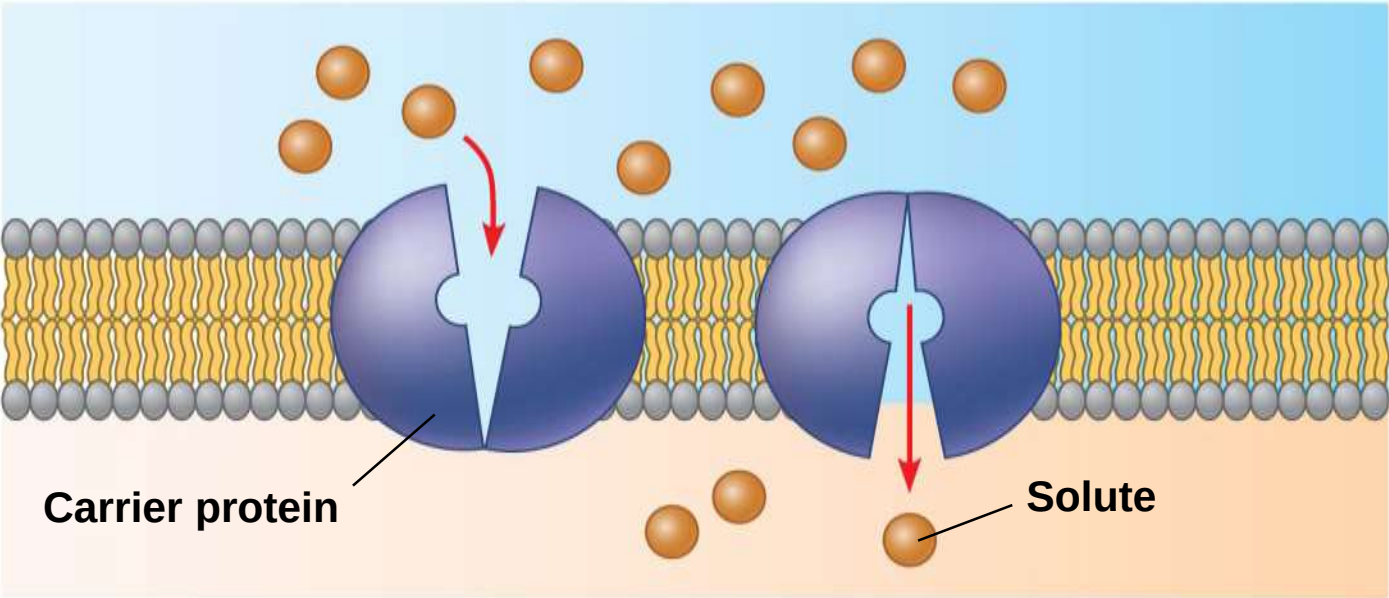
c) Ion channels facilitate the diffusion of ions

a) Some ion channels, called **gated channels**, open or close in response to a stimulus

Figure 7.14



(a) A channel protein



(b) A carrier protein

a) Carrier proteins undergo a subtle change in shape that translocates the solute-binding site across the membrane

Concept 8.4: Active transport uses energy to move solutes against their gradients

- a) Facilitated diffusion is still passive because the solute moves down its concentration gradient, and the transport requires no energy
- b) Some transport proteins, however, can move solutes against their concentration gradients

The Need for Energy in Active Transport

- a) **Active transport** moves substances against their concentration gradients
- b) Active transport requires energy, usually in the form of ATP
- c) Active transport is performed by specific proteins embedded in the membranes

- a) Active transport allows cells to maintain concentration gradients that differ from their surroundings
- b) The **sodium-potassium pump** is one type of active transport system

Figure 7.15

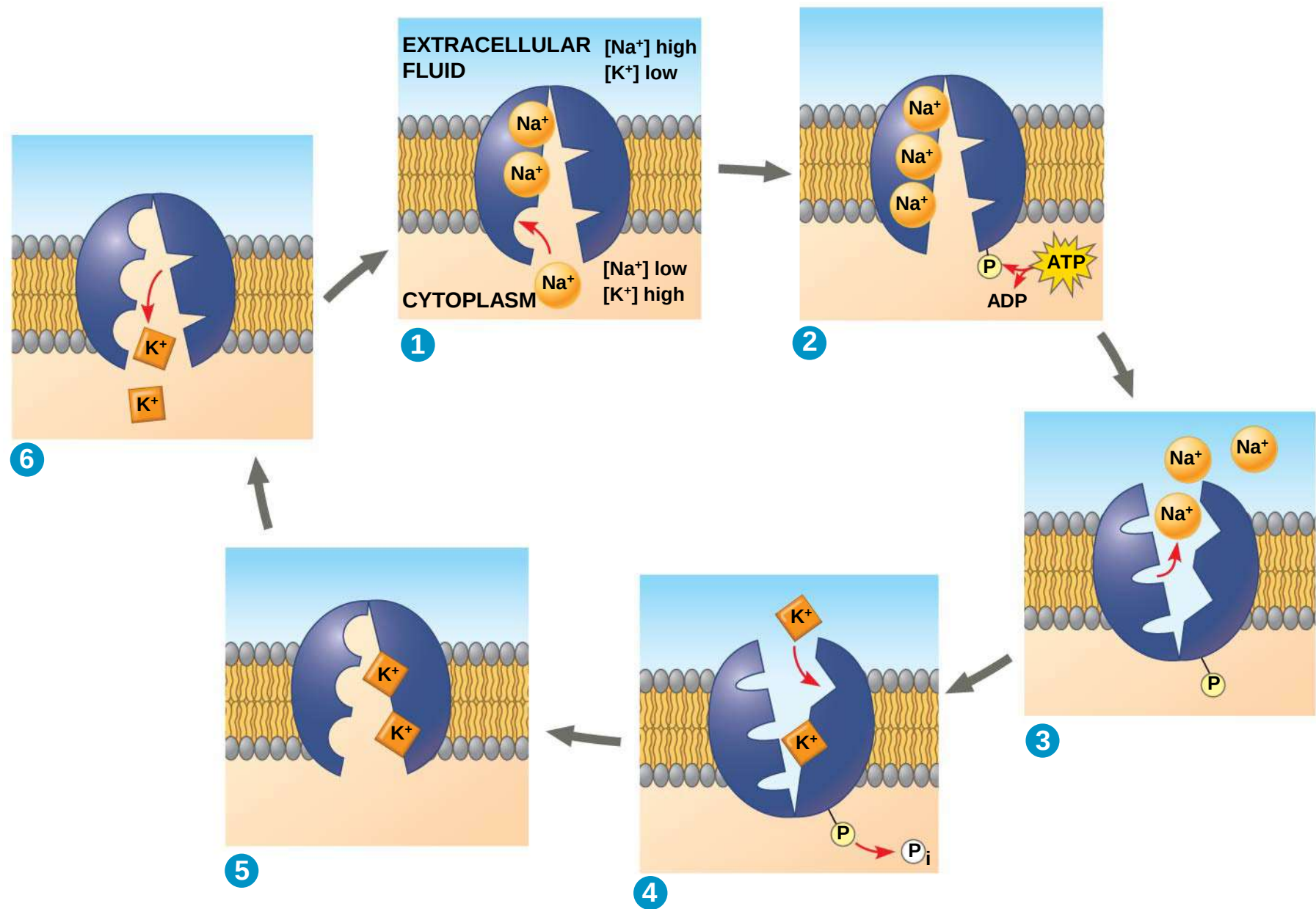
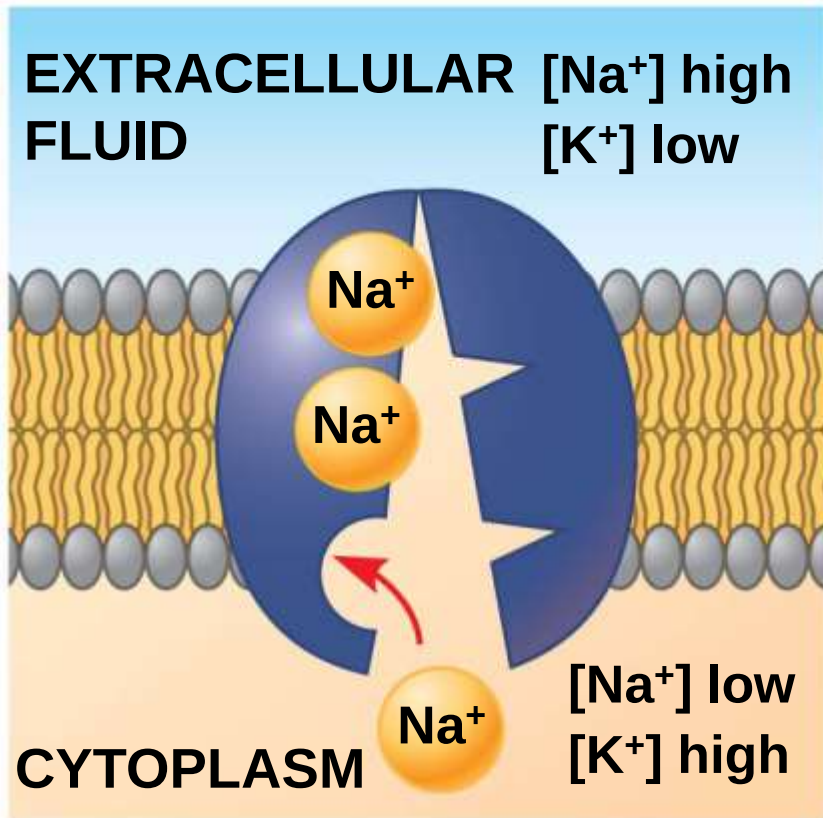
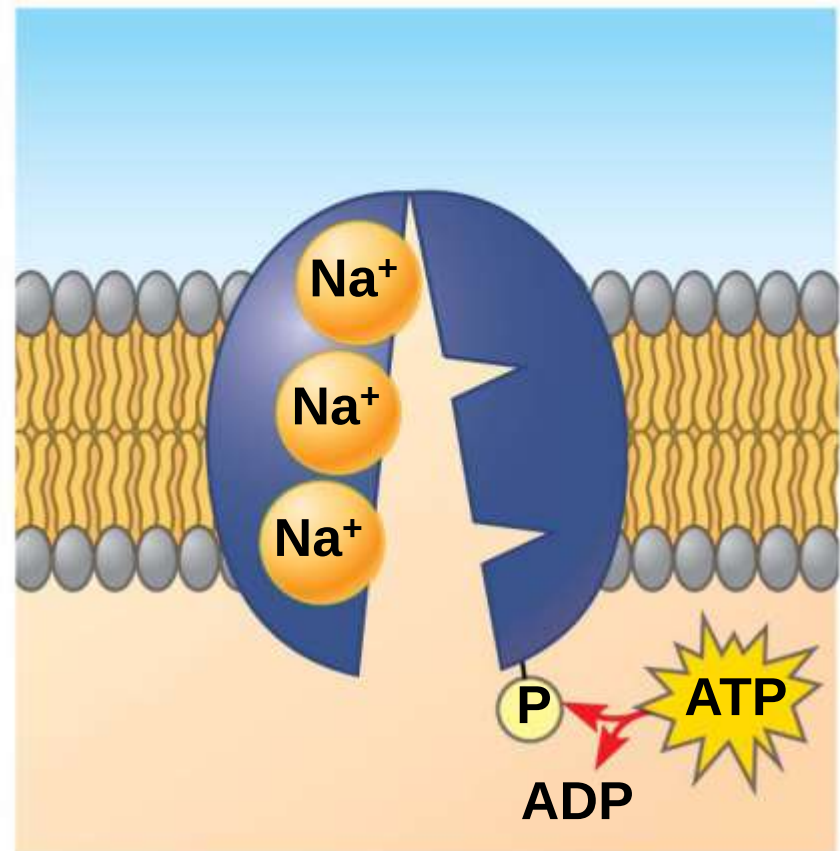


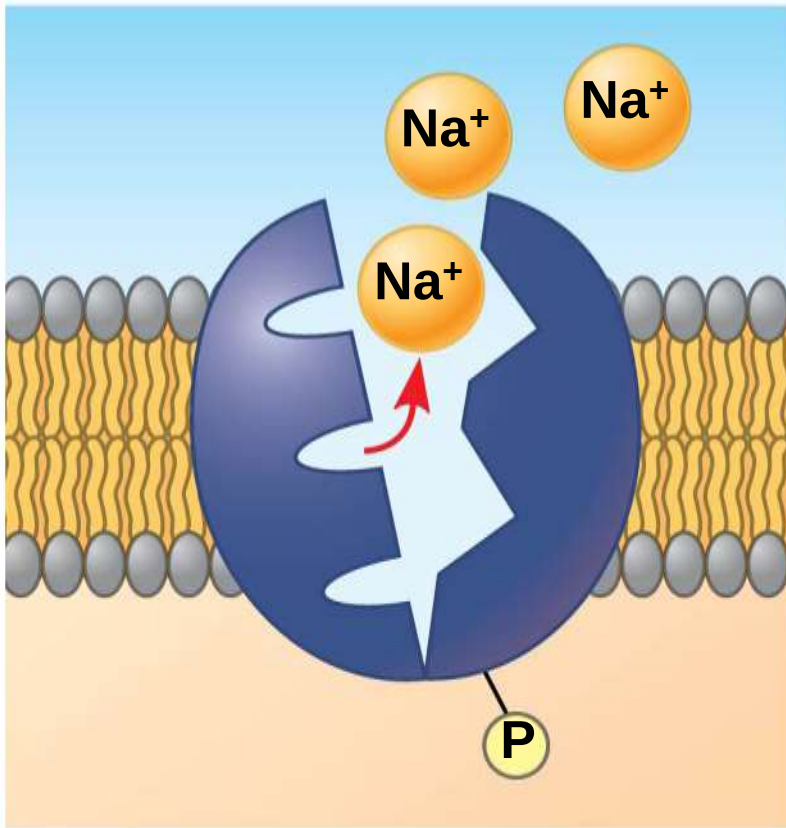
Figure 7.15a



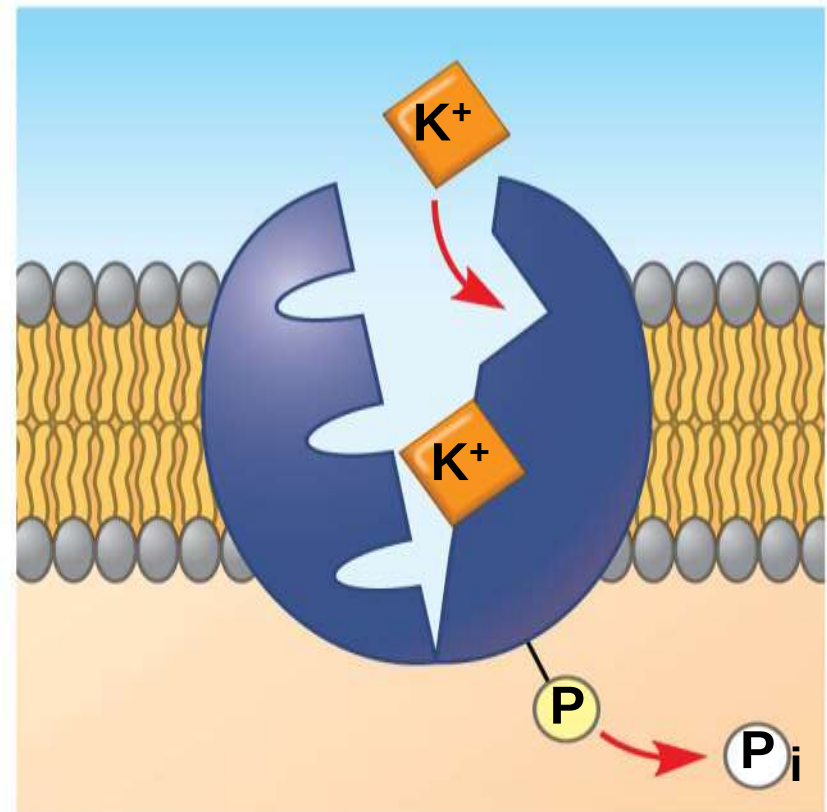
1 Cytoplasmic Na^+ binds to the sodium-potassium pump. The affinity for Na^+ is high when the protein has this shape.



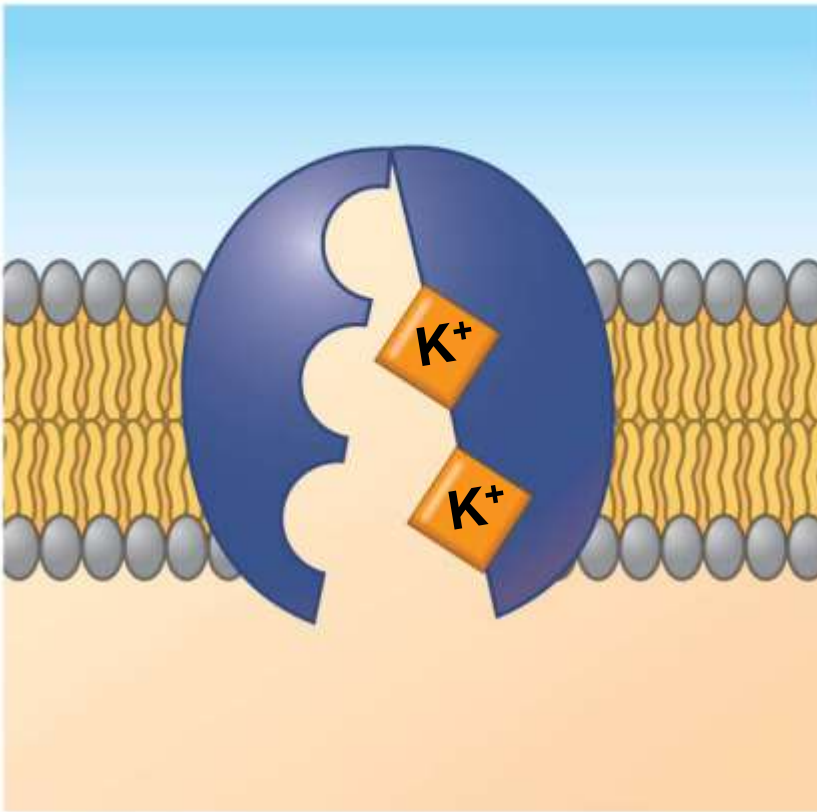
2 Na^+ binding stimulates phosphorylation by ATP.



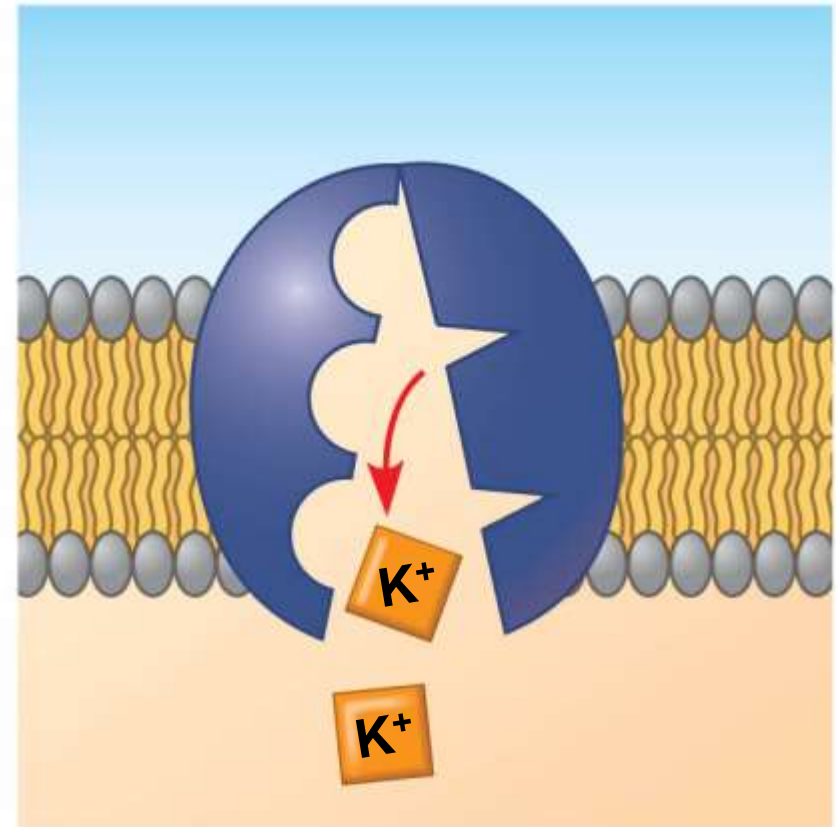
3 Phosphorylation leads to a change in protein shape, reducing its affinity for Na^+ , which is released outside.



4 The new shape has a high affinity for K^+ , which binds on the extracellular side and triggers release of the phosphate group.



5 Loss of the phosphate group restores the protein's original shape, which has a lower affinity for K^+ .



6 K^+ is released; affinity for Na^+ is high again, and the cycle repeats.

Animation: Active Transport

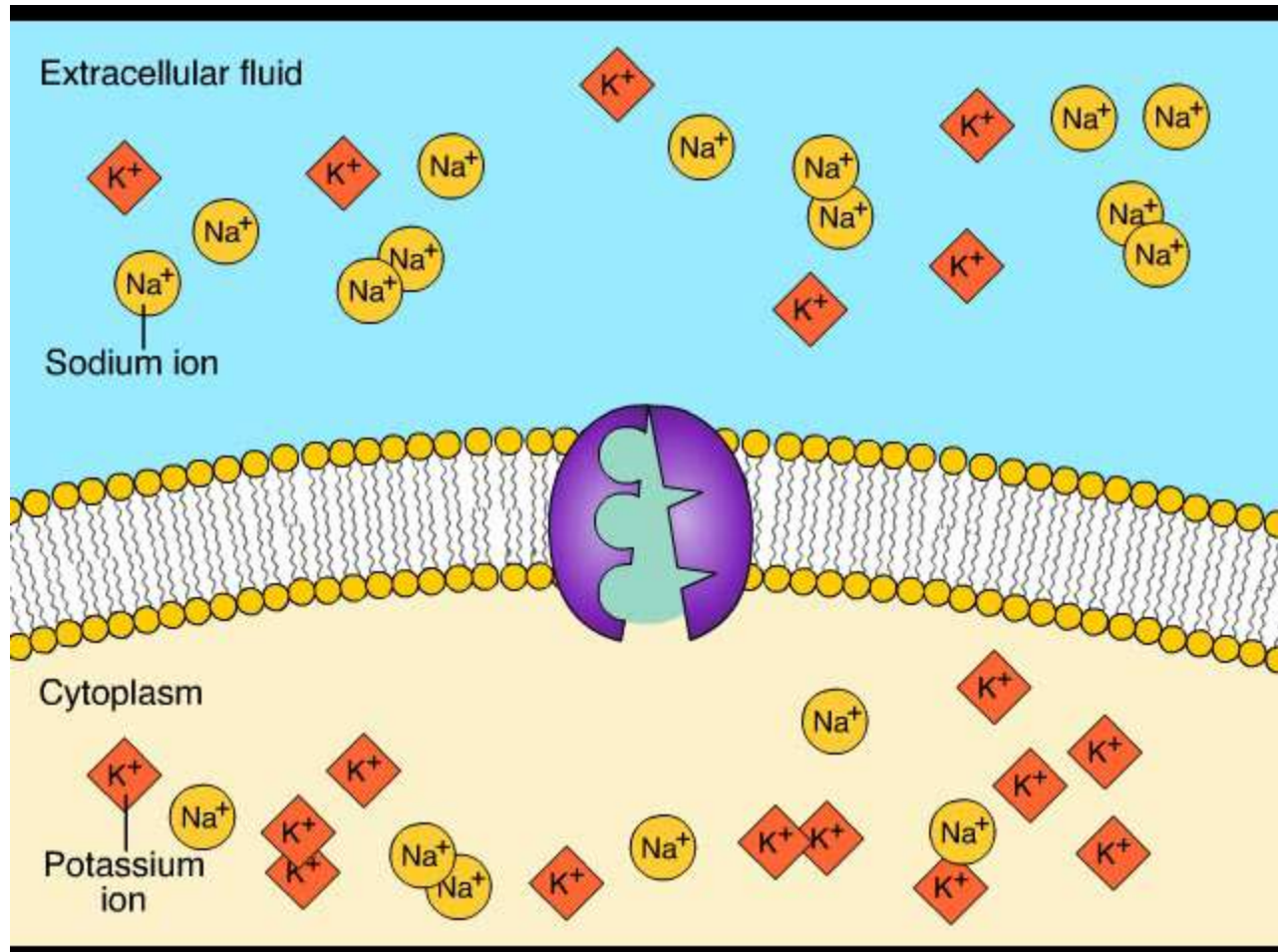
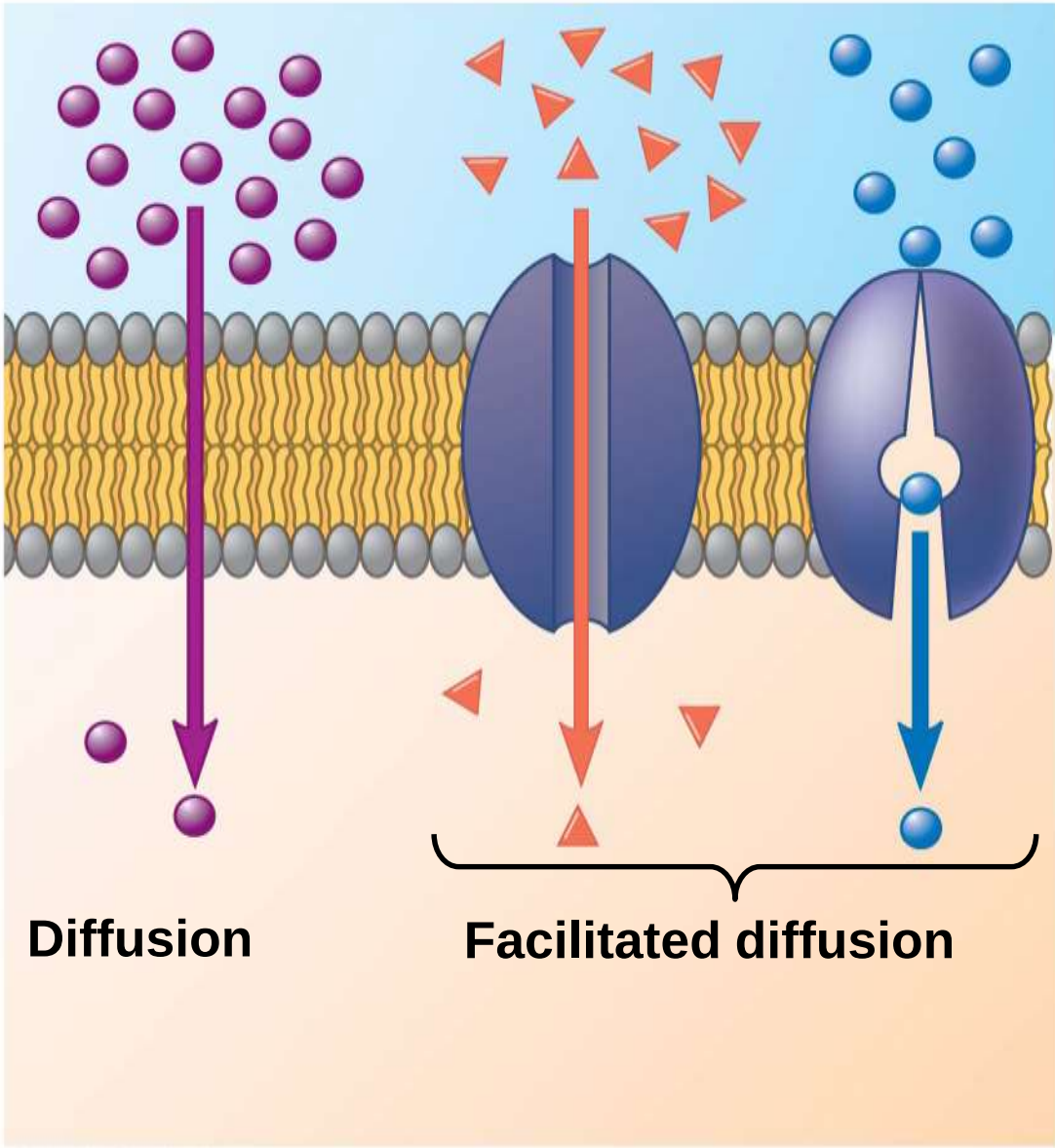
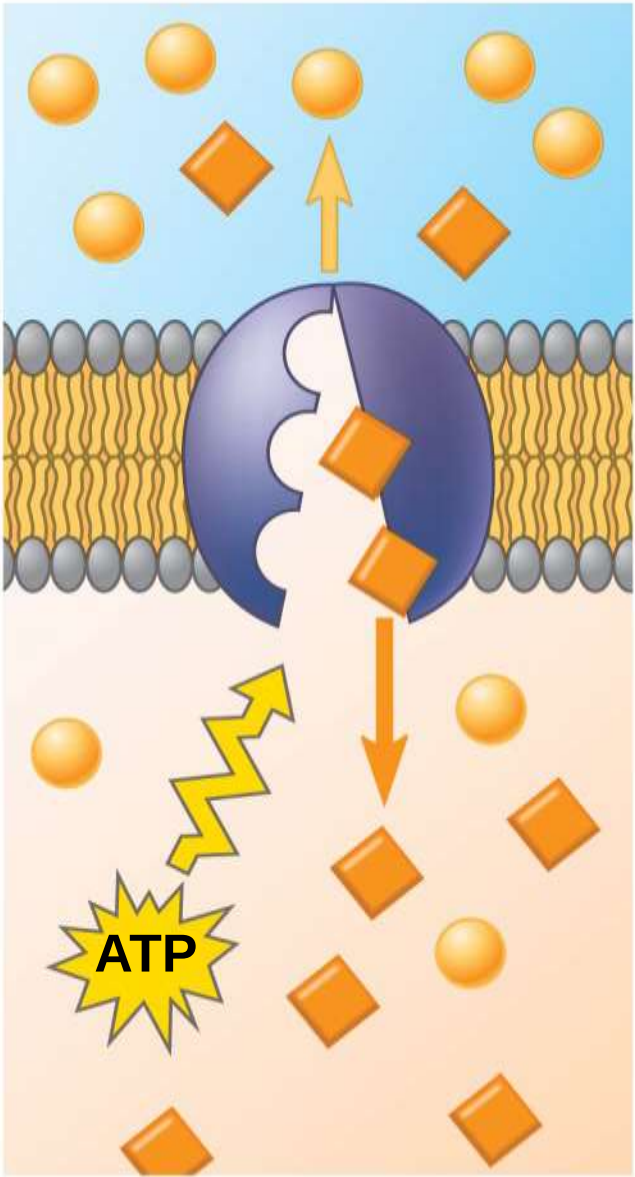


Figure 7.16

Passive transport



Active transport



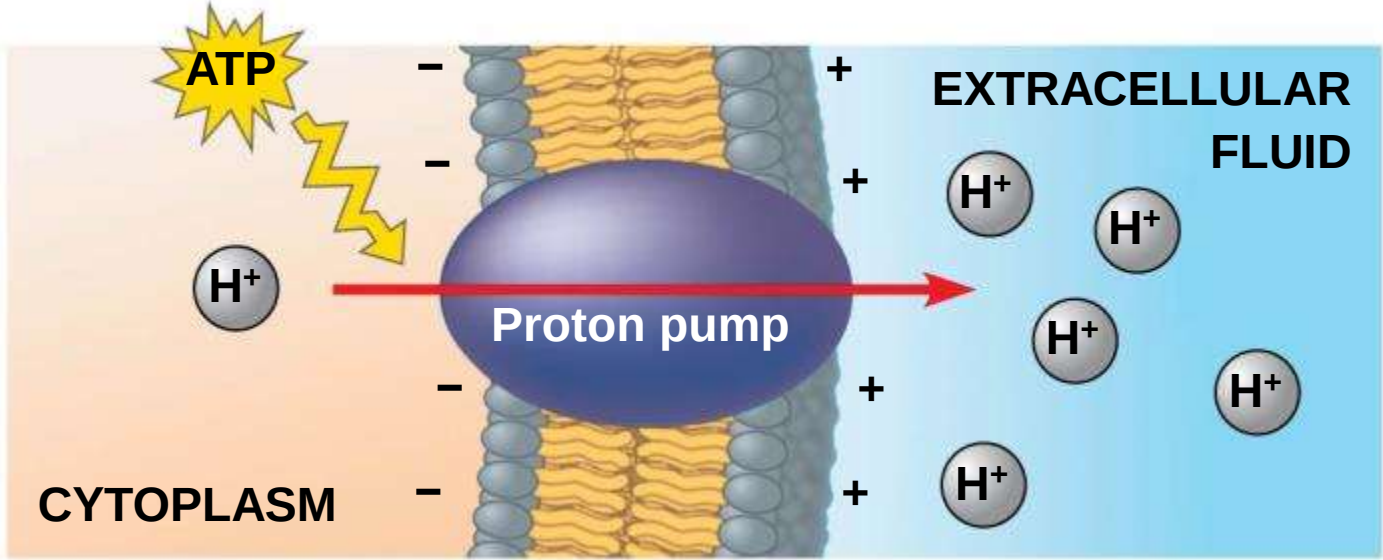
How Ion Pumps Maintain Membrane Potential

- a) **Membrane potential** is the voltage difference across a membrane
- b) Voltage is created by differences in the distribution of positive and negative ions across a membrane

- a) Two combined forces, collectively called the **electrochemical gradient**, drive the diffusion of ions across a membrane
 - a) A chemical force (the ion's concentration gradient)
 - b) An electrical force (the effect of the membrane potential on the ion's movement)

- a) An **electrogenic pump** is a transport protein that generates voltage across a membrane
- b) The sodium-potassium pump is the major electrogenic pump of animal cells
- c) The main electrogenic pump of plants, fungi, and bacteria is a **proton pump**
- d) Electrogenic pumps help store energy that can be used for cellular work

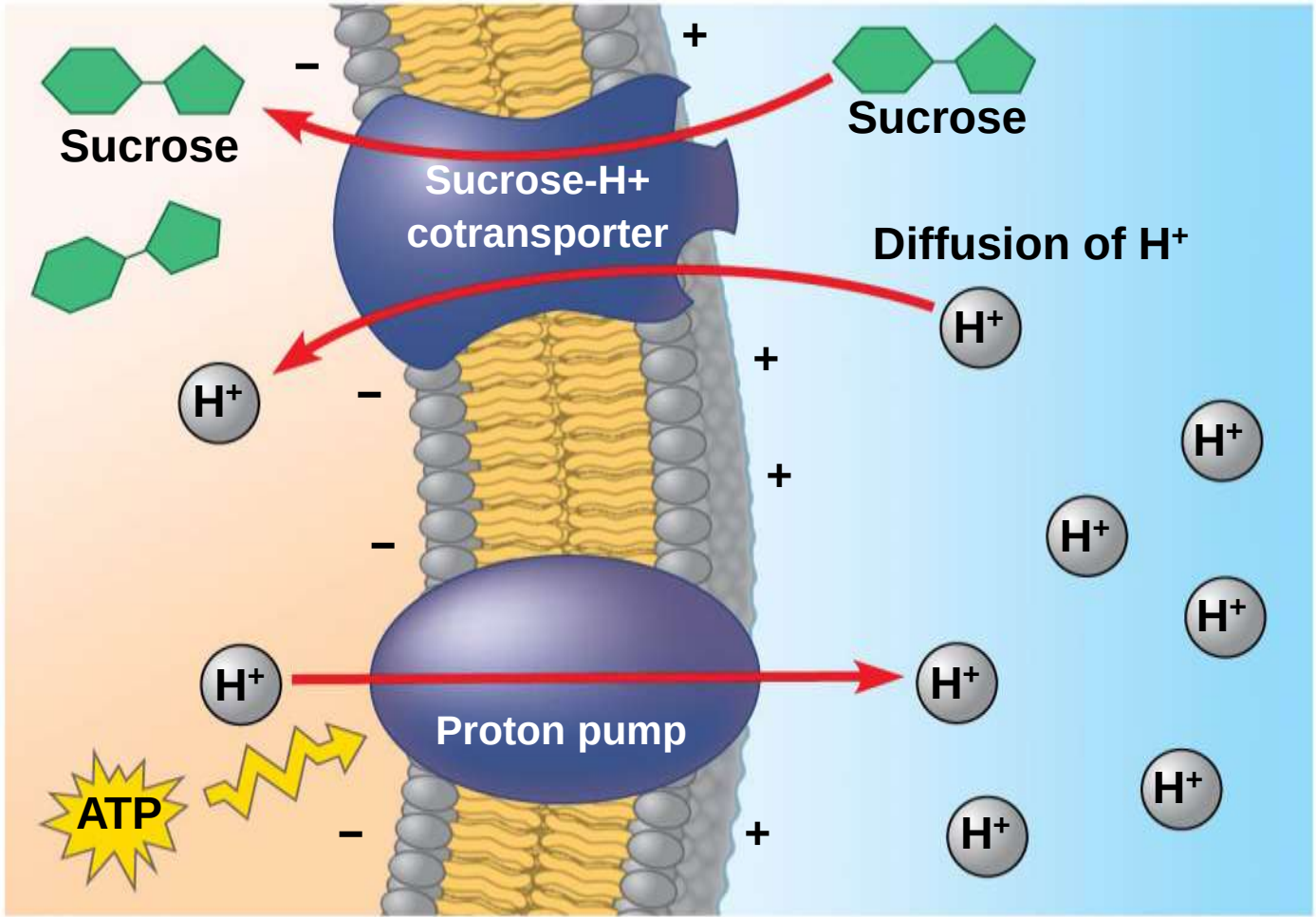
Figure 7.17



Cotransport: Coupled Transport by a Membrane Protein

- a) **Cotransport** occurs when active transport of a solute indirectly drives transport of other substances
- b) Plants commonly use the gradient of hydrogen ions generated by proton pumps to drive active transport of nutrients into the cell

Figure 7.18



Concept 7.5: Bulk transport across the plasma membrane occurs by exocytosis and endocytosis

- a) Small molecules and water enter or leave the cell through the lipid bilayer or via transport proteins
- b) Large molecules, such as polysaccharides and proteins, cross the membrane in bulk via vesicles
- c) Bulk transport requires energy

Exocytosis

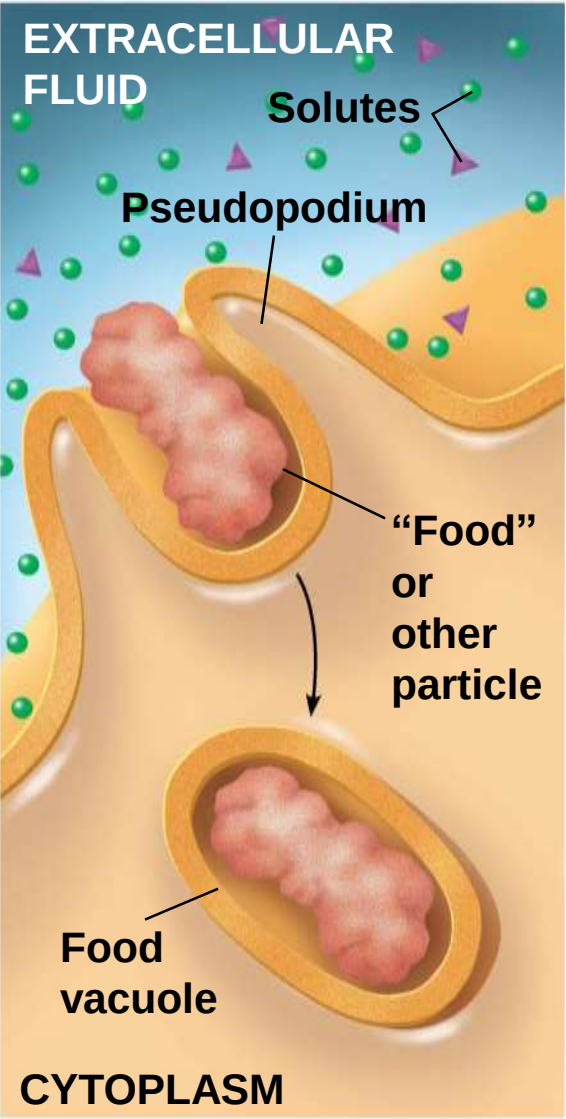
- a) In **exocytosis**, transport vesicles migrate to the membrane, fuse with it, and release their contents outside the cell
- b) Many secretory cells use exocytosis to export their products

Endocytosis

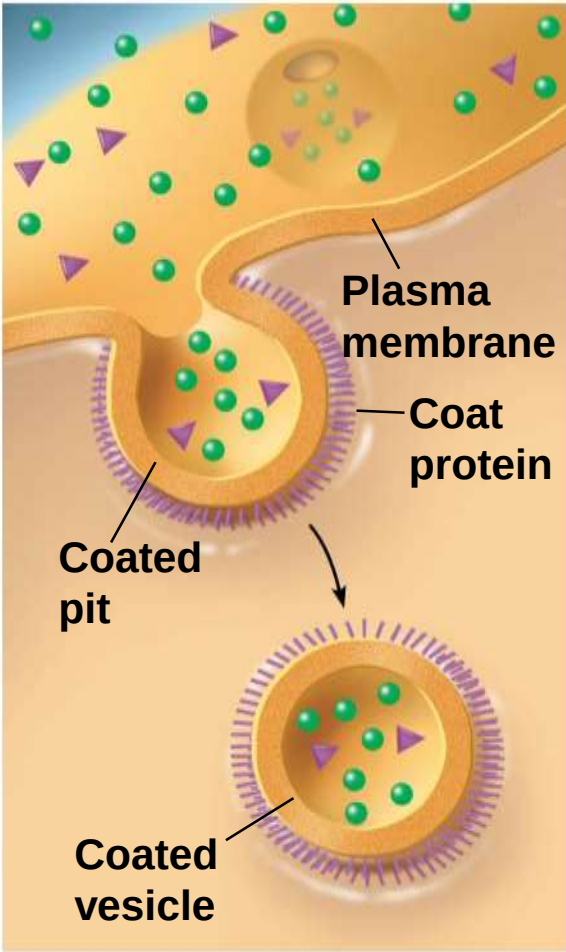
- a) In **endocytosis**, the cell takes in macromolecules by forming vesicles from the plasma membrane
- b) Endocytosis is a reversal of exocytosis, involving different proteins
- c) There are three types of endocytosis
 - a) Phagocytosis (“cellular eating”)
 - b) Pinocytosis (“cellular drinking”)
 - c) Receptor-mediated endocytosis

Figure 7.19

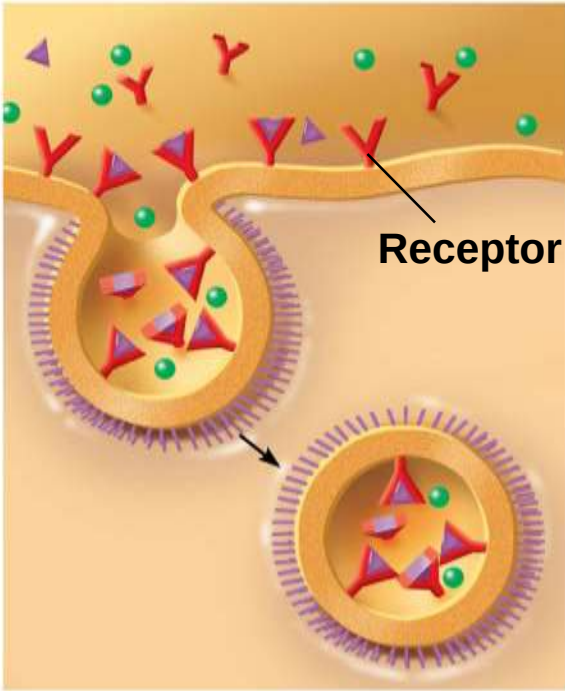
Phagocytosis



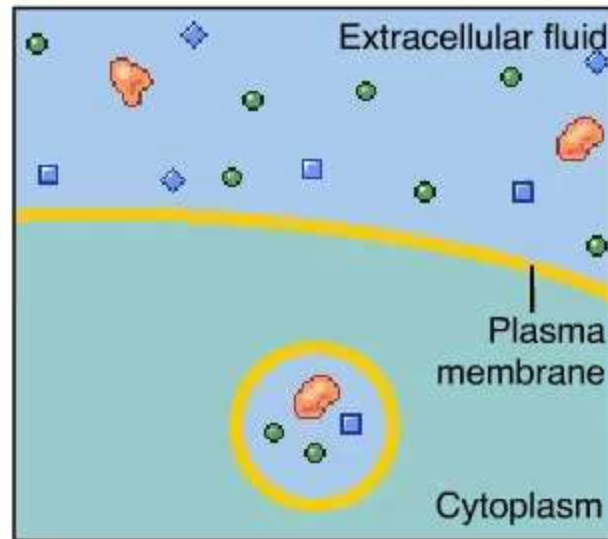
Pinocytosis



Receptor-Mediated Endocytosis

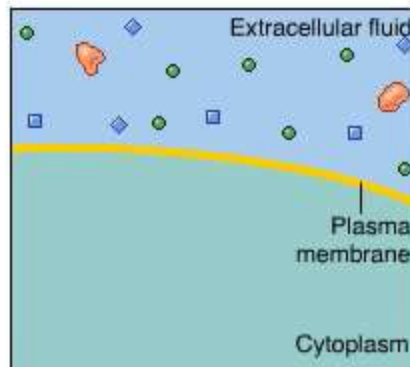


Animation: Exocytosis



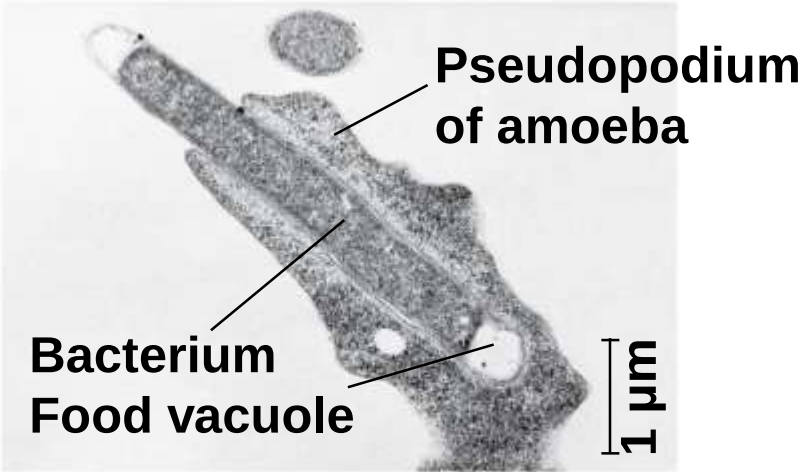
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Animation: Exocytosis and Endocytosis Introduction

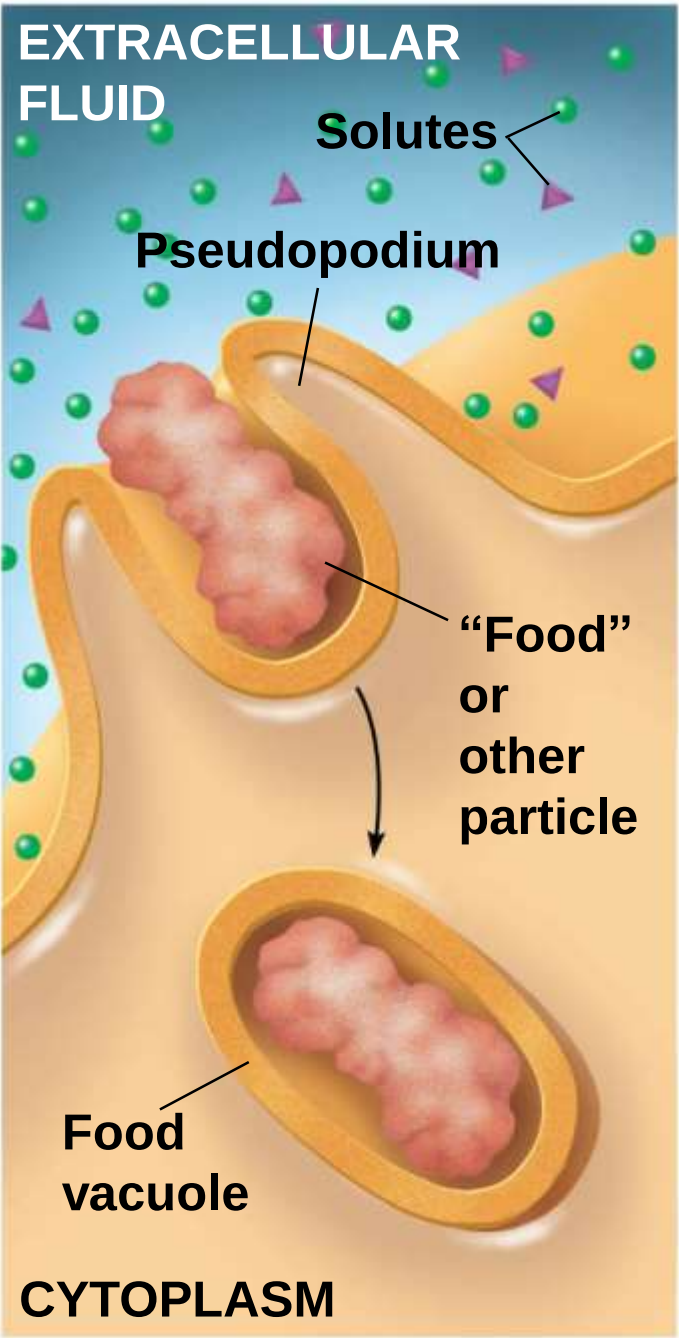


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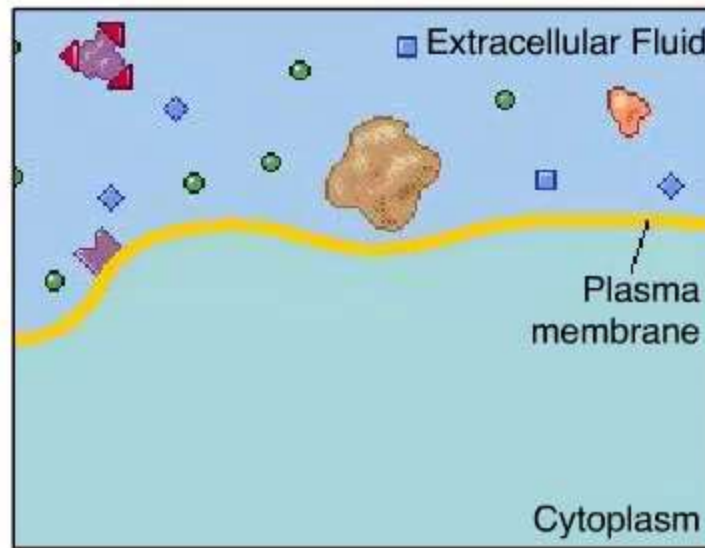
Phagocytosis



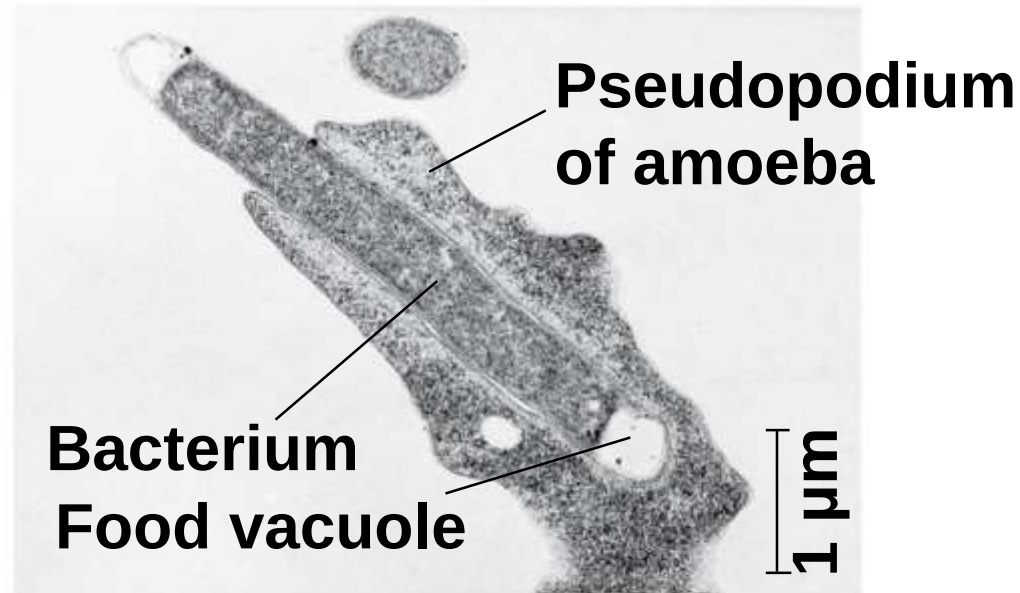
An amoeba engulfing a bacterium via phagocytosis (TEM)



Animation: Phagocytosis

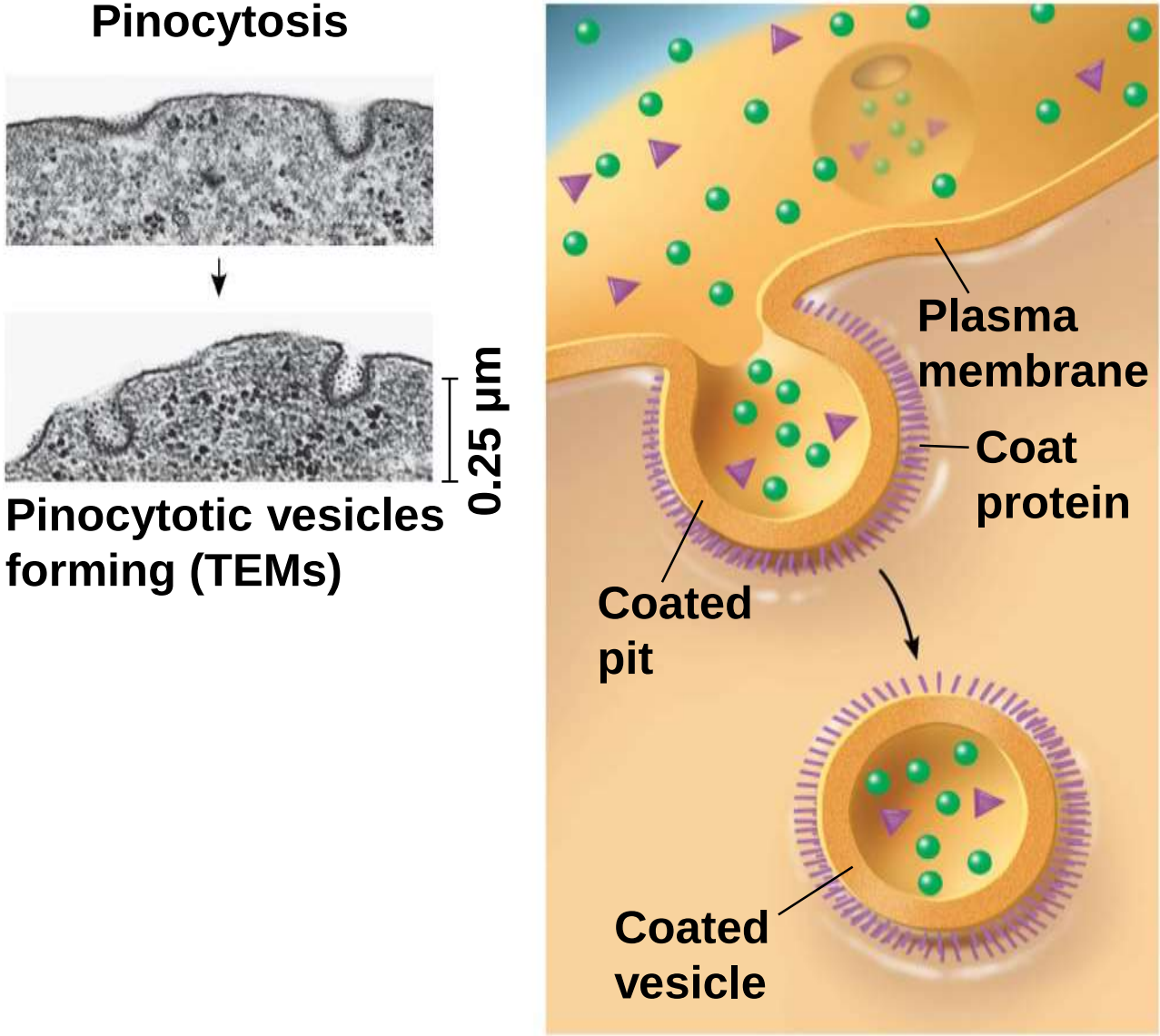


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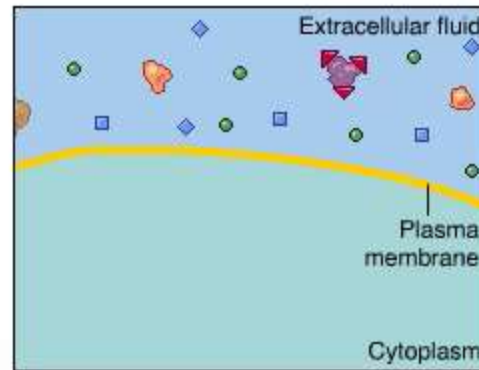


An amoeba engulfing a bacterium via phagocytosis (TEM)

Figure 7.19b

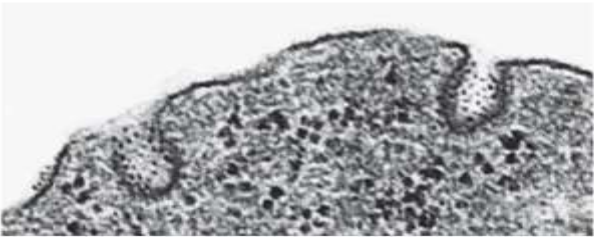
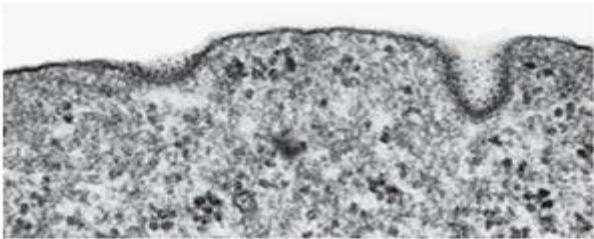


Animation: Pinocytosis



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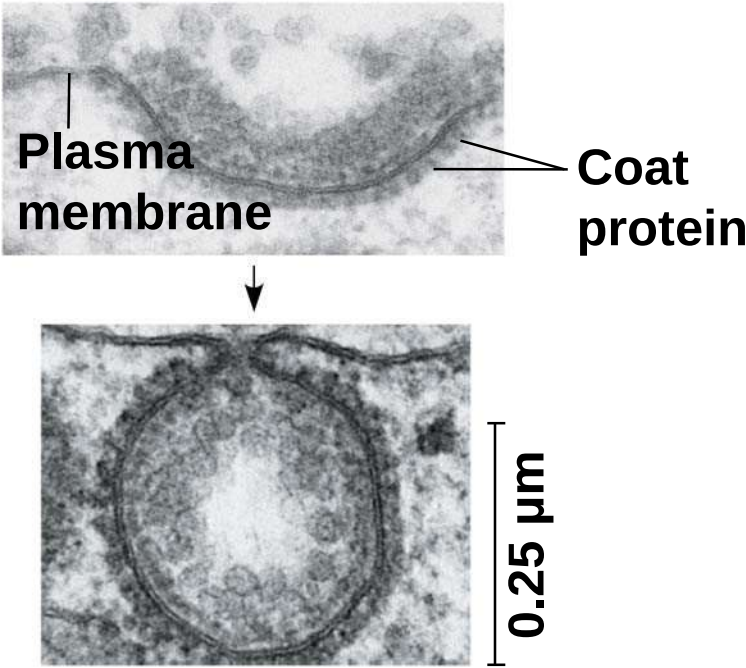
Figure 7.19ba



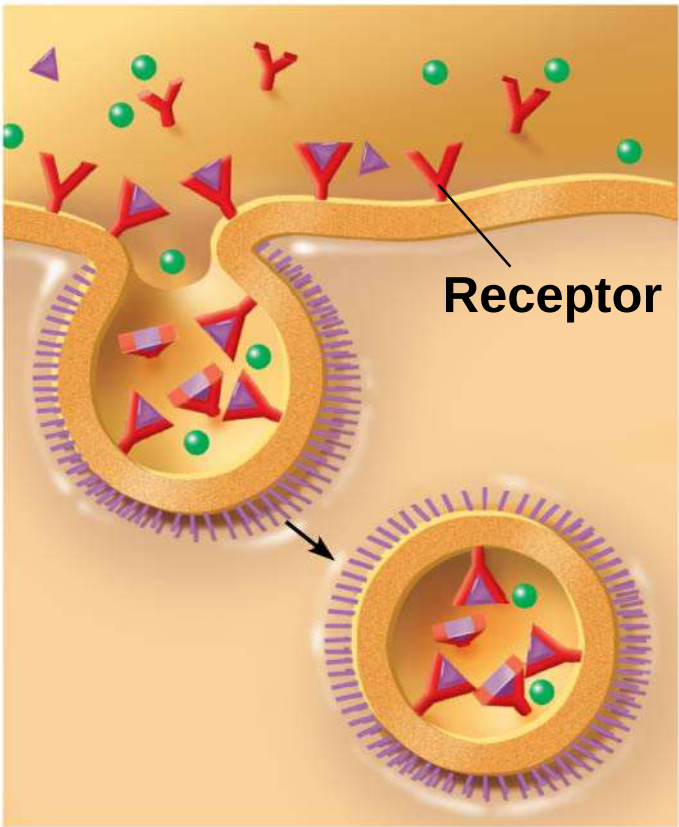
0.25 μm

**Pinocytotic vesicles
forming (TEMs)**

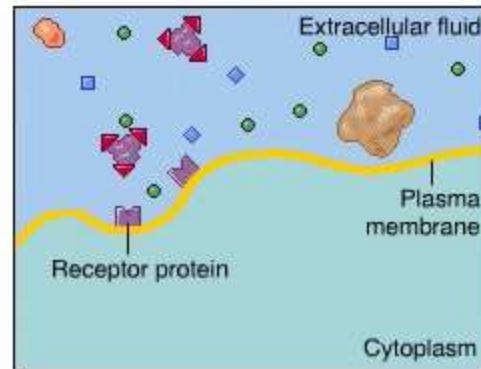
Receptor-Mediated
Endocytosis



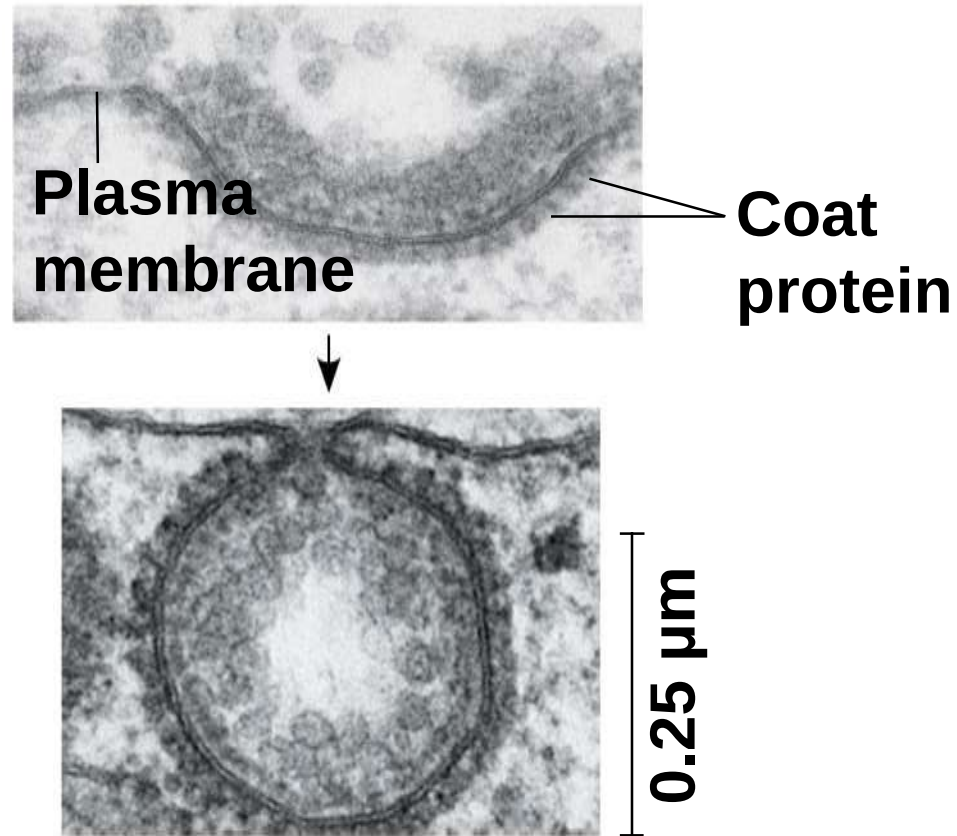
Top: A coated pit.
Bottom: A coated vesicle forming during receptor-mediated endocytosis (TEMs)



Animation: Receptor-Mediated Endocytosis



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Top: A coated pit.

Bottom: A coated vesicle forming during receptor-mediated endocytosis (TEMs)

- a) In **phagocytosis** a cell engulfs a particle in a vacuole
- b) The vacuole fuses with a lysosome to digest the particle

a) In **pinocytosis**, molecules dissolved in droplets are taken up when extracellular fluid is “gulped” into tiny vesicles

- a) In **receptor-mediated endocytosis**, binding of ligands to receptors triggers vesicle formation
- b) A **ligand** is any molecule that binds specifically to a receptor site of another molecule